Hype Cycle for Generative AI, 2025

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By: Arun Chandrasekaran, Leinar Ramos

Initiatives:Artificial Intelligence; Define the Future-State Architecture for the Enterprise; Deliver IT and Business Value Through Enterprise Architecture; Architect, Implement and Scale Data and Analytics Solutions; Generative AI Resource Center; Integrate Technology to Achieve Business Outcomes

Generative AI technologies and techniques are evolving at an unprecedented pace, matched only by the surrounding hype, which makes it challenging for leaders to navigate this dynamic landscape. AI leaders can leverage this Hype Cycle to identify the innovations most worth pursuing as they shape and execute their AI strategies.

Strategic Planning Assumption

By 2028, over 95% of enterprises will have used generative AI (GenAI) APIs or models, and/or deployed GenAI-enabled applications in production environments.

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Analysis

What You Need to Know

This Hype Cycle demystifies the core technologies underpinning the transformative generative AI (GenAI) trend. The technologies on the Hype Cycle for Generative AI, 2025 fall into four key areas:

- **GenAI models:** Pretrained AI models are the cornerstone of generative AI. The unprecedented pace of innovation in models has provided enormous choice and value to enterprises, although concerns about the models' environmental footprint, reliability and transparency remain.
- **AI engineering:** A growing ecosystem of GenAI tools and techniques enables organizations to build, customize and govern GenAI applications.
- **AI agents, applications and use cases:** Organizations are seeking to use AI agents to automate complex, multistep processes at scale, aiming to boost productivity, lower operational costs and improve the customer experience. Vendors are increasingly integrating AI into SaaS applications to deliver improved decision making and accelerated workflows. However, concerns regarding product quality and pricing remain prevalent.
- **Infrastructure and enablement techniques:** GenAI exists and is advancing through the combination of novel techniques and established AI practices. As AI becomes increasingly distributed, infrastructure innovations are evolving beyond traditional cloud environments to support edge computing, the Internet of Things (IoT) and on-device deployments.

AI leaders should use this Hype Cycle to identify GenAI innovations they can exploit that align with their organization's risk tolerance and strategic objectives in pursuit of maximizing potential rewards.

The Hype Cycle

The GenAI technology landscape consists of four key technology areas, which are detailed in the following sections.

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GenAI Models

Large language models (LLMs) are central to the GenAI revolution. Over the past year, they have rapidly evolved into multimodal systems, demonstrating significant advancements across image, speech and video domains. As retrieval-augmented generation (RAG) gains traction and popularity, embedding models have become essential for representing enterprise data within GenAI workflows.

Domain-specific models are being developed for a range of business functions like supply chain management. They are also gaining traction across sectors including healthcare, life sciences, legal, financial services and the public sector. Open GenAI models are rapidly advancing in terms of quality, speed of innovation and accessibility relative to their closed-source counterparts. However, ongoing debate surrounds the definition of "open source" in the context of GenAI, due to limited access, lack of transparency regarding training datasets and the prevalence of customized licenses (for more details, see Answering IT Leaders' Top 10 Questions on Open Generative AI Models). LLMs with better energy and resource efficiency have appeared over the past year that can run in local "edge" devices.

Large reasoning models are an evolution of AI models, capable of performing logical inference, complex problem solving and multistep reasoning. They are rapidly advancing through improved architectures and better training on structured and domain-specific data. The emergence of world models is pivotal to the future of AI. These models enable systems to simulate, comprehend and predict real-world environments. As a result, AI can exhibit more adaptive and goal-directed intelligence across a broad range of tasks.

Nevertheless, despite the rapid evolution of GenAI models in recent years, the achievement of artificial general intelligence — comparable to human-level intelligence — remains a distant aspiration. The following innovations pertain to the GenAI models area:

- Artificial general intelligence
- Domain-specific GenAI models
- Edge GenAI
- Embedding models
- Large language models
- Large reasoning model

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- Multimodal generative AI
- Open-source LLMs
- World models

AI Engineering

An expanding ecosystem of GenAI tools and techniques is enabling organizations to build, govern and customize GenAI-powered applications. The growing availability of model operationalization (ModelOps) tools has become increasingly critical as most enterprises are preparing to scale their GenAI pilots. Deployment of multiple models from various providers is becoming increasingly common, driving the need for sophisticated application orchestration frameworks.

Prompt engineering, retrieval-augmented generation (RAG), and reinforcement learning from human feedback are key techniques for training and effectively using GenAI models. RAG pipelines are being augmented with knowledge graphs (GraphRAG) to provide deeper grounding, reduce hallucinations and enable more contextual information retrieval. Mitigating disinformation through robust content validation and identity attestation is essential for regulatory compliance and for establishing user trust in AI systems. Ensuring trust, risk and security management (TRiSM) is important too, which has led to a huge growth of AI-specific tools. Tools such as AI gateways are designed to enforce the safe and efficient use of AI. Meanwhile model context protocol has emerged as an integration standard, enabling AI models to dynamically access and interpret relevant context. This ensures the delivery of accurate and consistent responses across applications.

The underlying data infrastructure for GenAI is also evolving with the emergence and convergence of specialized databases — such as vector databases with traditional databases. The following innovations pertain to the AI engineering area:

- AI gateways
- AI TRiSM
- Disinformation security
- GenAI application orchestration frameworks
- GraphRAG
- Model context protocol

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- ModelOps
- Prompt engineering
- Reinforcement learning from human feedback
- Retrieval-augmented generation
- Vector databases

AI Agents, Applications and Use Cases

GenAI virtual assistants, such as ChatGPT, have attracted much attention and are already further into the Trough of Disillusionment than any other innovation on this Hype Cycle. GenAI models and APIs are increasingly embedded into many GenAI-enabled enterprise applications, empowering organizations across diverse sectors to leverage them successfully. Software engineering has seen the strongest market adoption of these models and APIs. Synthetic data usage in the enterprise is moving toward the bottom of the Trough of Disillusionment, although its usage by the model providers to train LLMs continues to grow.

While AI chatbots function as assistive, passive tools, agentic AI, by contrast, represents an approach that focuses on enhancing the autonomy of AI systems — enabling them to plan, decide, act, and adapt in pursuit of defined goals. This will mark a fundamental shift in how organizations interact with AI systems and extract business value from them. AI agent development frameworks are growing in popularity, increasingly targeting both business users (no code) and enterprise developers to help enable creation of custom agents. Embodied AI empowers AI systems to interact with — and learn from — the physical world using sensors and actuators. This approach enables adaptive, context-aware intelligence and enhances performance in real-world environments.

The following innovations pertain to the AI agents, applications and use cases area:

- Agentic AI
- AI agent development frameworks
- AI-augmented software engineering
- Embodied AI
- GenAI virtual assistants

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- Generative AI-enabled applications
- Synthetic data

Infrastructure and Enablement Techniques

GenAI builds on a foundation of established AI techniques. Most AI foundation models are trained using self-supervised learning techniques on vast amounts of internet data.

Specialized infrastructure — such as generative AI workload accelerators — plays an important role in model training and the inference process. Most models will be trained and run in the cloud. However, as more open models become available, clients may train or run these models outside cloud environments — on-premises, edge and on devices. Demand will continue to grow for specialized AI chips and tools to run them efficiently and at lower cost. AI supercomputers are being rapidly built by hyperscalers to train more powerful and multimodal models of the future. Gartner advises organizations to adopt resource-efficient AI supercomputing infrastructures to maximize AI innovations, reduce infrastructure costs and future-proof their operations.

The following innovations pertain to the infrastructure and enablement techniques area:

- AI supercomputing
- Generative AI workload accelerators
- Self-supervised learning

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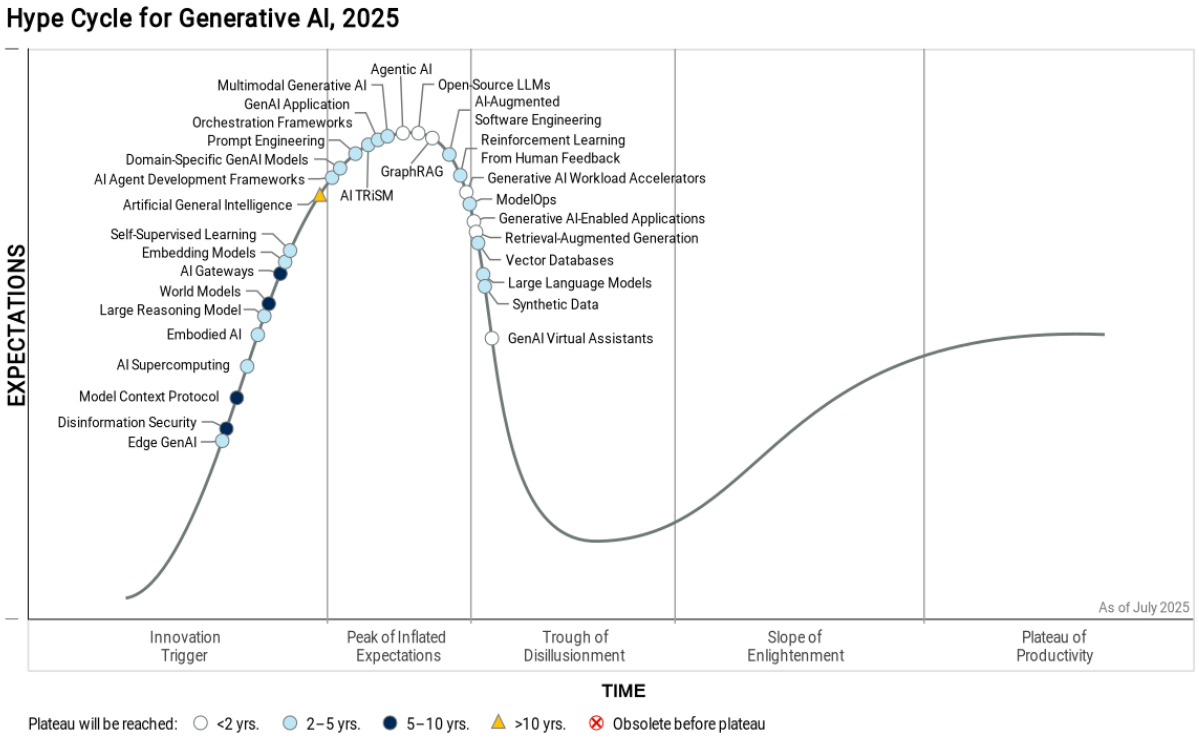
Figure 1: Hype Cycle for Generative AI, 2025



The Priority Matrix

While many innovations on this Hype Cycle are at the Innovation Trigger or Peak of Inflated Expectations, a number of technologies are sliding through the Trough of Disillusionment. Enterprises are increasingly aware of the challenges involved in moving from proof of concept (POC) to production. Many of these technologies, while steadily improving, still struggle to meet the elevated expectations, exacerbated by the intense hype surrounding generative AI.

The innovations that deserve particular attention within the two- to five-year period to mainstream adoption include AI agent development frameworks, edge GenAI and multimodal generative AI. Early adoption of these innovations can lead to significant competitive advantage and time-to-market benefits.

Several innovations remain five to 10 years away from mainstream adoption. Among these, disinformation security and world models present the greatest potential impact.

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Table 1: Priority Matrix for Generative AI, 2025

(Enlarged table in Appendix)

Benefit	Years to Mainstream Adoption			
↓	Less Than 2 Years ↓	2 to 5 Years ↓	5 to 10 Years ↓	More Than 10 Years ↓
Transformational	Agentic AI	AI Agent Development Frameworks AI-Augmented Software Engineering AI Supercomputing Embodied AI Large Language Models Large Reasoning Model ModelOps Multimodal Generative AI Self-Supervised Learning	World Models	Artificial General Intelligence
High	GenAI Virtual Assistants Generative AI-Enabled Applications Generative AI Workload Accelerators Open-Source LLMs Retrieval-Augmented Generation	AI TRiSM Domain-Specific GenAI Models Edge GenAI Embedding Models GenAI Application Orchestration Frameworks Prompt Engineering Reinforcement Learning From Human Feedback Synthetic Data Vector Databases	Disinformation Security Model Context Protocol	
Moderate	GraphRAG		AI Gateways	
Low				

Source: Gartner (July 2025)

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On the Rise

Disinformation Security

Analysis By: Akif Khan, Alfredo Ramirez IV, Apeksha Kaushik

Benefit Rating: High

Market Penetration: 1% to 5% of target audience

Maturity: Emerging

Definition:

Disinformation security is a suite of technologies and strategies that can address false and misleading content and narratives to help organizations discern trust, protect their brand and secure their online presence. Capabilities found within this category cover deepfake detection, impersonation prevention, reputation protection and narrative intelligence in addition to mitigation strategies based on countering narratives and grounding of information.

Why This Is Important

Disinformation deceives or misleads an audience with fabricated information that omits or maliciously alters critical context. Malicious actors target organizations to manipulate buyers, sway political opinion, diminish brand value, reduce share price and commit fraud. Today, tools are readily available to create fake content and disseminate it widely. Disinformation security aims to address these threats by combining several technologies for methodological discernment of trust.

Business Impact

Disinformation can be weaponized to target an organization or an individual leader. Campaigns can be put into action that will impact brand reputation, by degrading or distorting brand image, leading to loss of revenue. Targeted disinformation attacks have also been used to commit fraud against organizations. Disinformation impacts multiple business functions and teams, including executive leadership, security, public relations, marketing, finance and sales, which makes disinformation security an important part of an organization's security strategy.

Drivers

- External-threat actors target organizations with disinformation for ideological purposes, as part of cyber attacks and to commit fraud.

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- Rapid improvement and advancements in adversarial AI increasingly make discerning legitimacy of multimedia content more difficult.
- Deepfake videos flood social media with disinformation. Large language models, coupled with GenAI voice cloning, can quickly create convincing disinformation content “spoken” in almost any voice.
- The ability to mass-deliver disinformation via networks of bot social media accounts, and mass-customize its delivery, further escalates the threat.
- Security teams increasingly recognize that proactive defense necessitates looking beyond the organization’s public and private infrastructure. Many disinformation attacks are conducted using infrastructure not managed by the organization, from locations complicating proactive defense, removal and legal-team response.
- Marketing and public relations teams recognize the need to understand sentiment and any narratives that exist around an organization’s products, services, brands and executives.
- Organizations require digital watermarking, content verification and provable media systems to quickly and accurately identify legitimate versus manipulated and GenAI-created content.
- Organizations need to consistently and reliably know who they’re dealing with online; this includes the ability to accurately identify candidates in the hiring process, employees, contractors, businesses and customers.
- Organizations in regulated industries must comply with know-your-customer (KYC) requirements. The abuse of identity verification processes using GenAI-created content can expose organizations to serious regulatory sanctions.
- Account takeover or credential compromise are major security concerns for many organizations. Attack vectors are varied and can include using GenAI-created content to fool voice or face biometric authentication, or to trick identity verification processes used in account recovery workflows. Once authenticated as the perceived user, malicious actors can take nefarious actions such as planting ransomware, stealing intellectual property, theft of funds and spreading disinformation.

Obstacles

- Many organizations are still coming to grips with understanding and quantifying the risks posed by disinformation. Developing proactive security plans is still an emerging trend within the market.

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- Defensive technologies in this category are emerging; however, growing market hype can derail, confuse and complicate organizational strategies to combat disinformation.
- Emerging standards used to proactively label and identify GenAI-created content online, such as those from the Coalition for Content Provenance and Authenticity (C2PA), are still being finalized. Robust implementation that can be trusted will take time.
- New technologies are promising. However, maturation and effectiveness against sophisticated evolving adversaries remain to be seen. Many emerging solutions remain use-case-specific.
- Some organizations still do not understand that proactive security necessitates looking beyond their own four walls to develop effective defensive strategies.
- Detecting the domains in which some disinformation attacks occur, such as real-time impersonation of an organization's executive within a voice- or video-calling application, is a technical challenge as many platforms are walled-garden environments.

User Recommendations

- Adopt deepfake-detection technologies that combine a variety of AI techniques and practices, and digital forensic content verification capabilities to methodologically discern legitimate from artificial and fact from fiction in different use cases.
- Implement impersonation prevention technologies to evaluate user behaviors holistically across all devices and interactions to validate authentic actions and provide nonrepudiation. Ensure that biometric systems involving face or voice have liveness-detection capabilities deployed for employees, customers, suppliers and contractors.
- Conduct deepfake red-teaming. Most employees have never seen a deepfake of their leadership or someone they know, and this is a liability in the event of an attack against your organization.
- Protect the organization's reputation from external-threat actors by proactively monitoring the platforms that adversaries may be using. Most notably, this will involve narrative intelligence services focused on social media platforms.

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- Combat disinformation by forming cross-functional teams and strategies. Information security teams will continue to be responsible for management and administration of tools used to identify and detect disinformation impacting the organization, and the public relations and marketing communications teams should handle external announcements and communications strategy. This will require coordinated governance.

Sample Vendors

Alethea Communications Technologies; Blackbird Security; Breacher.ai; DeepTrust; Doppel; DuckDuckGoose; GetReal Security; LetsData; Reality Defender

Gartner Recommended Reading

Emerging Tech: Combating Deepfakes, Deception and Disinformation in Multimedia Content

Emerging Tech: Disinformation Security Techniques Must Be Used as Part of ITDR

How to Mitigate Deepfake Identity Impersonation Attacks

Emerging Tech: Disinformation Security Products Must Be Part of Cybersecurity

Edge GenAI

Analysis By: Chirag Dekate, Eric Goodness, Danielle Casey, Thomas Bittman

Benefit Rating: High

Market Penetration: 1% to 5% of target audience

Maturity: Embryonic

Definition:

Edge generative AI (GenAI) refers to the deployment of GenAI models at or near the edge in smartphones, PCs, IoT devices and edge servers. This innovation combines the rapid evolution of GenAI models and edge architectures, including agentic workflows, to deliver new breakthrough applications with significant business impact.

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Why This Is Important

Edge GenAI delivers more efficient, secure and contextualized GenAI models on local devices, resulting in faster functionality, better response times and real-time applications. It enables local processing of data and better privacy and security by minimizing data movement across the enterprise architecture. Edge GenAI also enables the delivery of GenAI models in disconnected or intermittently connected edge devices. Further RL-infused edge GenAI and agents enable decentralization and scalability of the core GenAI deployment architecture, delivering new breakthrough applications.

Business Impact

- Edge GenAI and edge agents enable real-time and contextualized applications in diverse industries, such as manufacturing, healthcare, finance, logistics and retail.
- Edge GenAI processes models closer to where data is created and managed, enabling enhanced privacy, security, efficiency and architectural simplicity. Agents further improve efficiency here by addressing process flows.
- Edge GenAI could amplify combinatorial impacts and create new immersive user experiences for new products, services and growth when integrated with adjacent innovative technologies such as robotics and XR.

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Drivers

- The need is emerging for immersive and interactive real-time analytics to power new business applications while maintaining stringent data protection.
- The need for highly contextualized and efficient GenAI models is increasing.
- GenAI is experiencing rapid evolution, exploration and adoption across enterprises, including at the edge.
- Maturing edge environments now enable collection of experiential data that can help address the pretraining scaling challenges that GenAI workflows struggle with. Edge integration for both data collection and for deployment of GenAI models can amplify efficiency gains.
- New advances in reinforcement learning, coupled with rapid evolution of feature-rich training and deployment of AI techniques, and middlewares that integrate physics simulation and AI are accelerating progress of AI at the edge. Examples of this include the physical AI technologies and stacks that NVIDIA, AMD and Intel are devising today that accelerate advancements of autonomous vehicle systems, robotics and other applications.
- Inference is key for deploying GenAI applications on edge devices that require specialized solutions to address limitations. Utilizing hardware accelerators like NPUs and software techniques such as quantization enables efficient performance of GenAI models, ensuring high-quality experiences within the constraints of edge environments.
- Combinatorial innovations exist across standardization of edge AI interfaces, specialized AI chips and GenAI model innovations.
- Demand has increased from regulated industries where GenAI models need to be run in air-gapped or edge environments.

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Obstacles

- **Limited value:** Real value delivery not living up to the extreme GenAI hype could have a cascading impact across the stack and inhibit evolution of the nascent edge GenAI market.
- **Cost:** Edge GenAI will almost always require high-performance compute architectures — especially in multimodal AI, throughput, concurrency and latency-sensitive scenarios — potentially driving up the cost of edge devices.
- **Rapid GenAI evolution:** The fast-evolving GenAI landscape could expose limitations of edge GenAI solutions that are overengineered in a dated GenAI context, resulting in faster obsolescence of premature solutions.
- **Lack of skills:** Lack of sufficient skills to engineer and deploy edge GenAI solutions could inhibit the evolution of this embryonic market.

User Recommendations

- Map potential use cases for edge GenAI and weigh them against measurable business value metrics, such as cost savings, top-line growth, customer satisfaction scores and improved physical safety or security, with a clear timeline for these goals.
- Audit your edge and IoT environments. Task your core team to identify gaps and risks and mitigate them before rolling out edge GenAI use cases.
- Track emerging developments in GenAI to ensure continuous improvement and progress of your edge GenAI strategy. Provide frequent feedback to your suppliers to ensure that they meet your evolving needs.

Sample Vendors

Amazon Web Services; AMD; Apple; Google; Hailo; Intel; Microsoft; Numenta; NVIDIA; Phison

Gartner Recommended Reading

Innovation Insight for Edge AI

Emerging Tech Impact Radar: Edge Artificial Intelligence

Market Guide for Edge Computing

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Model Context Protocol

Analysis By: Andrew Humphreys, David Pidsley

Benefit Rating: High

Market Penetration: Less than 1% of target audience

Maturity: Emerging

Definition:

Model context protocol (MCP) is an emerging standard to enable two-way communication between AI models and other applications and data sources. It provides a standardized way for applications to share contextual information with large language models (LLMs) and expose tools and capabilities to AI systems.

Why This Is Important

MCP was published by Anthropic in November 2024 and has been adopted by many vendors, including Amazon Web Services (AWS) and Microsoft. It supports LLM-based AI application development by simplifying connecting to sources of information and functionality. This has been challenging, as it required writing custom code or using APIs, which have not been designed for AI usage. MCP defines a standard way to access external data and tools in AI workflows, increasing consistency and interoperability.

Business Impact

MCP can lead to more context-aware AI systems and can reduce integration effort by standardizing how to give AI models access to contextually relevant and up-to-date multimodal data and multistructured analytics beyond their initial training data. However, MCP brings additional security concerns, as well as strategic questions about its usage. It also may be eclipsed by future standards that have greater openness and security.

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Drivers

- MCP extends the reach and simplifies implementing and maintaining LLM-based AI applications and agents. They can accomplish this because they can more easily be given access to new data sources or tools without retraining the model or significantly altering the core system, increasing the flexibility and agility of the AI solutions.
- Productivity and effectiveness for AI agent developers of LLM-based AI applications and agents is increased because connecting to data sources and tools is simplified. Developers can use MCP to integrate with multiple, diverse systems without needing to use system-specific approaches and tools, reducing the cost and time to integrate.
- AI service providers and multiple application, analytics, decision intelligence and middleware vendors are starting to offer prebuilt MCP-defined integration interfaces in their applications and MCP support in tools. This further helps grow the MCP community adoption and ecosystem and simplifies adoption. This can lead to increased value and innovation as teams can reuse predefined MCP interfaces for new use cases without needing to build new MCP interfaces or custom integration points.

Obstacles

- MCP adds new security risks, particularly related to authorization, to data and tools and raises concerns about data privacy and security. Ensuring that classified information is protected and that access is appropriately controlled is crucial.
- Many organizations already register tools and resources as part of their AI platform architectures. AI development tools such as AIXplain, Boomi AI Studio, Crew.AI, Langchain and SnapLogic already provide mechanisms to register resources to be accessible by LLM-based AI applications and agents.
- Implementing MCP servers can be complex, especially for legacy systems or diverse technology stacks and may lead to suboptimal architectures and governance. Integrating MCP into existing infrastructure may require significant effort and expertise.
- MCP is relatively new and is evolving rapidly, with limited backward compatibility. Early adopters may face challenges with keeping implementations up to date with the latest version of the standard.

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User Recommendations

- Investigate MCP clients to understand the benefits and risks, and compare these with the capabilities offered by your AI development platforms.
- Prototype MCP servers for your internal systems and data sources to facilitate streamlined access for AI applications developers.
- Avoid the highest risks associated with the current MCP standard by prohibiting remote MCP server implementations that are externally accessible. Any use of external MCP servers should be strictly controlled within a test environment to assess MCP benefits, rather than deploying them in production.
- Budget to address technical debt, ensuring your teams have the resources and time to monitor the standard's evolution and adapt their implementations accordingly, with the flexibility to abandon the standard if MCP loses its initial advantage.
- Strengthen security measures by requiring MCP servers to use HTTPS endpoints and implementing OAuth, while fortifying configurations to protect MCP client and server description files.

Sample Vendors

Anthropic; Axiom; Cloudflare; Composio; Google; Microsoft; OpenTools; Stripe; Zapier

Gartner Recommended Reading

Emerging Patterns for Building LLM-Based AI Agents

Market Guide for AI Application Development Platforms

Innovation Insight for the AI Agent Platform Landscape

Innovation Insight: Model Context Protocol

AI Supercomputing

Analysis By: Chirag Dekate, Sushovan Mukhopadhyay, Soyeb Barot

Benefit Rating: Transformational

Market Penetration: 5% to 20% of target audience

Maturity: Adolescent

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Definition:

AI supercomputing is a rapidly evolving domain where purpose-designed supercomputing systems combine state-of-the-art innovations in computational accelerators, specialized software, high-speed networks and performance-optimized storage to create an integrated platform that accelerates training and inference. It is also used for deployment of complex, computationally intensive AI models such as generative AI (GenAI) models and multiagent systems.

Why This Is Important

Purpose-designed AI supercomputers that offer extreme scale parallelism are crucial to driving innovation in advanced GenAI, data analytics and simulation environments. AI supercomputing is important because it enables processing of extreme scale datasets that are crucial to developing and applying innovative GenAI models. As GenAI evolves beyond large language models (LLMs) into more complex domains, including multimodality, the computational capability required can only be addressed using AI supercomputers.

Business Impact

AI supercomputing transforms businesses by:

- Powering enterprises to deliver game-changing AI by customizing GenAI models to their organizational data and application context.
- Enabling AI model innovators to develop and deliver new breakthrough GenAI models, including domain-specific models, different size GenAI models and multimodal models.
- Accelerating large-scale inference of GenAI models and driving value creation in both everyday AI and game-changing AI.

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Drivers

- **Massive datasets:** Leadership-class GenAI models rely on ingestion of internet-scale datasets for training. As the model mix evolves beyond LLMs to multimodal models, the resulting growth of data (both structured and unstructured) will drive the need for advanced AI supercomputing resources.
- **Algorithmic complexity:** GenAI today comprises a broad mix of different size models, domain-specific models and multimodal models. All the models rely on extreme scale parallelism offered by advanced AI supercomputers for both training and inference.
- **Accelerated adoption of GenAI:** Practical applications of GenAI across industries, especially game-changing AI, will require speed and accuracy that only AI supercomputers can provide.
- **Regional regulatory requirements:** Most of the early AI supercomputing capacity is clustered in the U.S. and Western Europe. The combined impact of data residency requirements and the accelerated pace of global adoption of GenAI will drive the need for delivering AI supercomputing everywhere.
- **Competition and investment:** Extreme hype and the promise of transformative impact is driving intense competition globally to create new, innovative GenAI models that broaden GenAI's practical applicability. Cloud service providers, hyperscalers and specialized hosting and colocation providers are investing billions of dollars in building advanced AI supercomputing foundations to address this demand.

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Obstacles

- **Supply chain risks:** More than 92% of advanced semiconductor manufacturing that powers AI supercomputers is sourced from TSMC. Geopolitical disruptions or natural disasters (such as earthquakes and typhoons) that disrupt AI supercomputing supply chains could pose significant risks for the market.
- **Energy constraints:** Graphics processing unit (GPU)-based AI supercomputing technologies require multi-gigawatt scale data centers that enterprises may not be able to address due to limitations of their existing facilities or their local utility provider.
- **Component challenges:** GPU-based AI supercomputing technologies integrate complex sets of components from high-bandwidth memory stacks to transformers that downshift power to GPU cores. Supply chain disruptions in these components will create challenges for growth of AI supercomputing.
- **Lack of value delivery from GenAI:** If the extreme hype in GenAI doesn't translate to broad-based value creation, it could pose obstacles to continued progress of AI supercomputing.

User Recommendations

- **Choose cloud over on-premises:** Cloud-based AI supercomputing offers access to latest innovations in hardware and software beyond GPUs. Use them to take advantage of elastic scale of prebuilt AI stacks. Use cloud-based resources when possible. When no alternatives exist, devise on-premises AI supercomputing strategies.
- **Adopt use-case-driven AI supercomputing:** Focus on identifying use cases that demand extreme computational power, such as GenAI training, retrieval-augmented generation, fine-tuning, and inference and deployment of GenAI models. Use cloud-based AI supercomputing for scoping AI supercomputing resources to derisk and rightsize your investments.
- **Select the right architecture:** AI supercomputers based on GPUs and the alternative AI application-specific integrated circuits available via leading cloud providers and innovative AI supercomputing vendors can deliver better lower cost, performance, accuracy and scalability. Not all models require GPU acceleration.

Sample Vendors

Amazon; AMD; Google; Groq; IBM; Intel; Meta; Microsoft; NVIDIA; SambaNova Systems

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Gartner Recommended Reading

An Overview of AI Platform Offerings

Market Guide for Specialty Cloud Providers

Embodied AI

Analysis By: Pieter den Hamer

Benefit Rating: Transformational

Market Penetration: Less than 1% of target audience

Maturity: Embryonic

Definition:

Embodied AI is based on the view that intelligence and embodiment in a certain context are inextricably linked — one shapes the other. It is an approach where a physical or virtual AI agent's models are trained and co-engineered with its embodiment: the user interface, sensors, appearance, actuators or other capabilities required to perceive and interact with a specific, real or simulated environment. This enables more robust, resilient and adaptive execution of intelligent tasks.

Why This Is Important

Embodied AI aims to create AI agents that can act autonomously or augment humans in practical, dynamic contexts — much more so than current AI, including abstract large language and "reasoning" models with limited reliability and effectiveness in decision making and action taking. This is achieved through active perception and adaptive behavior, orchestrated by an AI agent's intelligence that is in symbiosis with the capabilities and constraints of the AI agent's host or body in a certain environment.

Business Impact

Embodied AI paves the way toward more robust, trustworthy, adaptive and actionable AI, widening its applicability and value creation. This is particularly the case where there is a need for more practical know-how, physical common sense, social and emotional intelligence, and a greater resilience to deal with the dynamics and unexpected events in real-world or virtual environments. Example use cases include autonomous vehicles and smart robots, but also virtual assistants or gaming characters.

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Drivers

- Recent advances in GenAI and agentic AI are impressive, yet AI still has significant limitations, particularly with respect to its reliability in dealing with the dynamics and complexity of reality.
- Advances in realistic 3D/4D simulations, virtual/augmented/mixed reality and gaming. Combined with reinforcement learning for adaptive behavior training, this allows the co-evolving of baseline versions of both embodiment and intelligence of AI agents, before further deploying and improving them in a real environment, be it physical or virtual.
- Emerging approaches include world models, physics-informed or first-principles AI (representing, among others, the laws of physics or engineering heuristics), adaptive AI (learning during operations), emotion AI (understanding and expressing feelings in a social context), composite AI (e.g., using neurosymbolic AI for spatiotemporal reasoning) and causal AI (representing cause-and-effect relations).
- Innovation is ongoing in sensor technology, robotics engineering and, for example, new materials for more natural mechanics and haptic interfaces (relevant for embodied AI in physical contexts).
- Scientific insights about intelligence are evolving; intelligence is no longer seen as a centralized brain-only concept. Cognitive traits like perception, emotion, reasoning and behavior are often distributed and co-evolved in multiple parts of the body.
- Investments are being made in research to develop future artificial general intelligence, for which embodied AI is increasingly seen as a critical step, based on the view that intelligence is inseparable from its operational entity that interacts with the environment. This means it is not abstracted from but grounded in reality by design, holding the promise of providing intrinsic meaning or semantics to its knowledge representations and “native” common sense.

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Obstacles

- The world is a very complex, unpredictable and even chaotic place. That is why the development of realistic simulations, effective robotics and — for example — truly autonomous cars has proven to be elusive.
- Real-world interaction requires real-time, highly responsive AI, even with limited energy and compute resources (e.g., on mobile or edge devices). However, more lightweight and energy-efficient AI are not easily achievable.
- Embodied AI holds the promise of more autonomous AI. Unfortunately, this may not only facilitate benevolent but also malevolent use. Effective regulation and risk management for responsible AI are, however, not a given.
- AI embodiments can be — depending on the use case — unnecessarily humanoid in their design, bringing in additional complexity and challenges.
- Embodied AI requires multidisciplinary collaboration between experts in areas as diverse as machine learning, GUI design and mechanical engineering.

User Recommendations

- Identify use cases that may benefit from applying embodied AI, both in more virtual domains, such as online customer interaction or knowledge worker augmentation, and in more physical domains, such as manufacturing, logistics, healthcare or facility management.
- Explore the value that embodied AI can add by reducing the limitations of current AI in terms of better interpretation of, for example, physical constraints in a warehouse or cultural norms in client interaction. This may result in increased safety or decreased bias in the use of AI, respectively.
- Extend the mindset of how AI agents should be developed or trained. Move from a modeling-only approach toward one that considers how intelligence can be a synergy between AI models and the design of the agent's embodiment. This could, for example, relate to the facial expression of virtual agents, or the coordination of movement in physical agents.

Sample Vendors

DEEP Robotics; Figure AI; Guerrilla Games; Intrinsic; NVIDIA; Qualcomm; Sereact; Toshiba; Unitree; Wayve

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Gartner Recommended Reading

Innovation Insight: AI Agents

Innovation Insight: AI Simulation

Hype Cycle for Mobile Robots and Drones, 2024

Top Strategic Technology Trends for 2025: Polyfunctional Robots

Large Reasoning Model

Analysis By: Tong Zhang, Leinar Ramos

Benefit Rating: Transformational

Market Penetration: 1% to 5% of target audience

Maturity: Emerging

Definition:

A large reasoning model (LRM) is an advanced type of large language or multimodal model designed to address complex problems that necessitate deliberate, multistep logical processes. It boosts problem-solving capabilities by eliciting reasoning chains, utilizing reinforcement learning and conducting extensive multistep attempts during inference to refine and enhance outputs.

Why This Is Important

Standard LLMs or large multimodal models provide instant response for all types of inputs. However, they lack the capability to solve complex problems by generating intermediate steps, exploring potential solution paths, verifying the correctness of intermediate results and backtracking from dead ends before coming to the solution. LRMs are designed to spend more effort tackling complex tasks beyond simple knowledge extraction, such as planning, advanced math problems and challenging coding tasks. As LRMs recognize hidden patterns among those complex problems, their reasoning capability potentially can be generalized for other tasks.

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Business Impact

- LRMs are specifically designed to handle complex tasks that go beyond mere fact retrieval. They incorporate structured reasoning, iterative reflection and verification mechanisms, enabling them to potentially perform at high levels on business tasks that require logical reasoning and problem-solving skills. Examples include supply chain optimization, complex financial risk assessment and healthcare diagnostics and treatment planning.
- LRMs can improve the reliability of LLM-enabled solutions via generation of long chains of thoughts, exploration of different problem-solving strategies and self-verification.
- LRMs have the potential to replace standard LLMs as the core AI model of various AI agents and automation tools, enhancing their planning capabilities and leading to more accurate and reliable solutions.

Drivers

- Diminishing returns from scaling up pretraining in standard LLMs highlight the need for a paradigm shift toward LRMs. Allocating extra computing resources in LRMs during inference allows models to spend more time refining their responses. Techniques such as generating chains of thought, revising answers, employing external verifiers, backtracking, and sampling multiple times to select the best answer significantly improve model performance.
- Increased interest in AI agents drives the need for LRMs, which are widely expected to become a core AI agent component, to improve their ability to tackle problems involving complex business processes.
- Large action models (LAMs) are foundation models trained and optimized to identify and generate an action or set of actions that can be used to impact a target environment to meet a goal. LAMs can augment reasoning models by improving their action-oriented capabilities. This integration allows LRMs to not only process information but also to execute tasks, bridging the gap between understanding and action.
- Use of model distillation drives LRM adoption. Training distilled smaller models from LRMs achieves performance that surpasses that of small language models while requiring fewer computing resources than the original LRMs. This lowers inference and training costs, making LRMs more accessible to organizations.

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Obstacles

- LRMs have higher cost and latency than standard LLMs, limiting their adoption in many enterprise use cases.
- Little evidence suggests LRMs possess a genuine understanding of concepts or the causal relationships inherent in tasks, even though they excel at recognizing patterns within data. They interpret patterns in historical data without grasping why those patterns exist.
- LRMs are trained to induce or emulate more structured, explicit, step-by-step reasoning capabilities, but their black-box nature still presents a big challenge for businesses seeking reliable and explainable solutions.
- Enterprises find effectively measuring the reasoning capabilities of LRMs to be difficult, which hampers adoption, and current benchmarks face limitations such as data contamination and narrow scope.
- Models can get stuck in repetitive thinking patterns, consuming excessive tokens and time. The result is inefficient resource usage and latency.
- Doubts remain about LRMs' ability to generalize their reasoning capability to a wider array of use cases, although they have shown improved performance in specific domains like mathematics and science.
- LRMs may act inappropriately in certain circumstances to gain rewards, for example, cheating on coding tests, misleading users or avoiding difficult problems.

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User Recommendations

- Pilot LRMs in areas where complex capabilities for decision making, strategic planning or problem solving are required and LLMs failed to achieve the desired outcome.
- Provide employees foundational training about LRM definition and normal use cases.
- Experiment with LRMs as the core model for AI agent solutions, replacing traditional LLMs to check whether performance can be improved.
- Create nuanced evaluation methods for LRMs including quality of outcome, token consumption and inferencing latency.
- Acknowledge the inherent black-box nature of LRMs and mandate appropriate levels of validation, testing and human oversight to ensure deep interpretability.

Sample Vendors

Alibaba Cloud; Anthropic; Baidu; DeepSeek; Google; MoonShot AI; OpenAI; xAI; Zhipu AI

World Models

Analysis By: Mike Fang, Nick Ingelbrecht, Sushovan Mukhopadhyay

Benefit Rating: Transformational

Market Penetration: Less than 1% of target audience

Maturity: Emerging

Definition:

World models are learned abstract representations of an environment. They enable AI systems to make predictions via simulating potential future states and helping to understand the consequences of the actions taken.

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Why This Is Important

AI systems struggle to function effectively in physical environments due to challenges such as safety concerns, restricted data coverage, limited adaptivity to novel situations, and the absence of cause-and-effect reasoning capabilities. World models are fundamental for efficiently forming representations based on the environment, constructing plans, and simulating events and their outcomes. They offer insights into potential effects of actions in the environment, which are crucial for AI agents.

Business Impact

- By capturing the underlying principles and regularities of the environment, world models can enable the simulation and anticipation of future states and outcomes based on current conditions and actions. This allows AI systems to acquire knowledge, refine their models and apply learned insights to new situations for informed decision making, even in unfamiliar contexts.
- World models could provide AI applications with a controlled environment for experimentation, allowing researchers and developers to explore different strategies, algorithms and policies before deploying them in the real world.

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Drivers

- World models have applicability across various sectors, from film production to autonomous vehicles and robotics. Their ability to enable simulation and anticipate complex interactions makes them invaluable tools for AI agents to achieve innovation and efficiency in diverse fields.
- World models empower AI to perform more sophisticated prediction and planning tasks, moving beyond mere pattern recognition in observed data. By simulating and understanding the dynamics of environments, AI can better handle uncertainty or missing information and therefore make informed decisions that account for future possibilities and contingencies.
- These models can be used to enhance the realism and credibility of generated video content by incorporating physical laws and constraints. This ensures that the produced visuals adhere to the principles of physics, resulting in more believable and immersive experiences.
- Trained on extensive multimodal datasets derived from robots functioning in real-world scenarios while combining first-principle AI capabilities, world models can guide robots in object manipulation and interaction with their environments.
- World models assist embodied AI in comprehending associations, counterfactuals, interactions and modeling the dynamics of the world. They go beyond summarizing observed content by efficiently simulating potential scenarios to predict outcomes, thereby enabling the selection of optimal actions.
- Research from control theory and cognitive science, such as Joint-Embedding Predictive Architecture (JEPA), has highlighted alternative approaches to construct world models.

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Obstacles

- Simulating real-world environments and inferring causal relationships is one of the most challenging domains of AI, and therefore building world models. Counterfactual reasoning requires simulating hypothetical causes and predicting outcomes, but current models are limited.
- Simulating physical laws is challenging for world models, especially in capturing real-world physical rules. Existing synthetic video generation models like Sora simulate phenomena like object motion and light reflections, but struggle with complex physical effects like fluid dynamics and aerodynamics, lacking accuracy and consistency.
- Techniques supporting world models mainly interpolate data, not extrapolate. The real world has many uncertainties, making world models difficult to build.
- Unlike humans, world models need a very large amount of situational and contextual combinations, leading to high computational costs. Additionally, acquiring real-world data faces challenges like public availability, security and privacy issues.

User Recommendations

- Avoid relying solely on GenAI techniques for world modeling as a solution for every use case; instead, leverage a broad array of methods from both physical AI and cognitive science to create a more comprehensive and effective model.
- Utilize extensive multimodal datasets, including sensory inputs like images and sounds, to train or customize world models for better contextual understanding and decision making across diverse scenarios.
- Implement strategies to mitigate bias and ethical issues in world models, ensuring fair and unbiased decision making in AI systems.
- Manage expectations around these techniques, as they are still surrounded by hype. Begin by piloting them in more focused or non-mission-critical use cases through mini world models limited to constrained environments (like in game situations).

Sample Vendors

Covariant; Decart; Google; Meta; NVIDIA; OpenAI; VERSES; World Labs

Gartner Recommended Reading

Innovation Insight: Causal AI

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Innovation Insight: AI Simulation

Emerging Tech: Generative AI's Path to Revolutionary Computer Vision Products

AI Gateways

Analysis By: Andrew Humphreys, Mark O'Neill

Benefit Rating: Moderate

Market Penetration: 1% to 5% of target audience

Maturity: Embryonic

Definition:

AI gateways, also known as LLM routers or multi-LLM gateways, manage and secure connections to AI providers. Some AI gateways are based on established API gateway products, since REST APIs are the primary mechanism for connecting to AI providers, while others are purpose-built. AI gateways can be used to apply security, such as protection of API keys issued by AI providers, multi-LLM routing, cost visibility, and data privacy scanning to an organization's AI usage.

Why This Is Important

AI gateways manage interactions between applications and AI models by providing security, governance, observability and cost management. This reduces the risk of unexpected costs incurred from AI providers, and prevents private data in API traffic from being compromised or misused. An AI gateway may also enable an organization to manage usage of multiple AI models, from different providers, rather than focusing the organization on one provider.

Business Impact

As the volume and scale of generative AI and other types of AI projects have increased, organizations require greater control over their use of AI providers. AI gateways provide runtime traffic management between the organization and their AI providers. The gateways can help implement and manage prompt-based policy controls, track AI service use and costs, route across multiple LLMs, and manage access to AI subscriptions, including protecting API keys issued by AI providers.

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Drivers

- **Control access costs:** Pricing models for AI services tend to be token-based, which presents a risk for businesses as the costs associated with use of these services can accrue rapidly, and requires special metering ability beyond an API gateway's typical transaction-based metering. Organizations are looking to control access costs, and AI gateways can help by caching responses to limit duplicate calls and tracking and controlling access to the services.

- **Enable visibility of use of AI services:** AI gateways provide a way to get greater visibility into the use of AI APIs across the organization by leveraging API management observability and analytics features for API usage.

- **Optimize access to AI engines:** AI gateways can be configured to provide a single API in front of multiple different AI providers, such as multiple LLMs. This means that developers can access multiple AI services using the same API.

- **Enforce security on AI interactions:** AI trust, risk and security management (AI TRiSM) ensures AI model governance, trustworthiness, fairness, reliability, robustness, efficacy and data protection. AI gateways address aspects of AI TRiSM, including model governance and reliability. In particular, protection of API keys issued by AI providers is vital. If an attacker gets access to an organization's API keys for an AI provider, they may access private data and run up large usage bills. AI gateways can be used to protect API keys issued by AI providers.

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Obstacles

- AI gateways are new and largely unproven and in the rapidly evolving AI environment new requirements emerge frequently; for example, to be able to manage Model Context Protocol endpoints. Conducting a thorough evaluation before committing to a vendor is a requirement.
- Latency is a concern for any intermediary, and especially in AI where end users may notice any delay in response time. Unlike traditional API latency, the important metrics here are time-to-first token and token per seconds (or token throughput). An AI gateway may help users optimize these two metrics.
- A single point of failure is also a concern if an AI gateway is deployed without redundancy in place.
- Caching of AI API responses is ineffective and complex, since different AI models may use different prompts. Determining whether natural language requests are requesting the same information is difficult, particularly when dealing with a large context window.

User Recommendations

- Consider using an AI gateway if you are looking to implement multiple LLM use cases or running AI applications at scale, as they provide centralized control, security, and cost management that is focused on AI-specific requirements that are not supported by standard API gateways.
- AI gateways are a relatively new category, and the features that are supported differ between vendor offerings. Conduct a thorough evaluation to evaluate the specific features and capabilities of the AI gateway you're considering to ensure that it meets your requirements.
- For simpler scenarios, such as rate-limiting of AI APIs, evaluate your existing API gateway's capabilities to act as an AI gateway, to see if they already or can be upgraded, to support your requirements.
- Ask your AI providers whether they can provide similar functionality, which would remove the requirement for a separate AI gateway.

Sample Vendors

Aguru; Cloudflare; lunar.dev; Martian; Portkey; Radiant Technology

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Gartner Recommended Reading

AI Pushes API Management Providers to Reimagine Their Products

Market Guide for API Gateways

How to Calculate Business Value and Cost for Generative AI Use Cases

10 Best Practices for Optimizing Generative AI Costs

Innovation Insight: AI Gateways

Embedding Models

Analysis By: Arun Chandrasekaran

Benefit Rating: High

Market Penetration: 20% to 50% of target audience

Maturity: Adolescent

Definition:

Embedding models are machine learning models used to represent high-dimensional data (like text or images) into vector embeddings. These vectors are represented across the multidimensional space, making it easier to perform tasks like semantic search, similarity comparison, clustering and classification on the data, often as part of a retrieval-augmented generation (RAG) architecture.

Why This Is Important

Embedding models are fundamental to AI because they enable machines to understand, represent and process complex data efficiently. In the context of RAG, embedding models play a crucial role by presenting relevant context from a vast amount of information, which is then used to generate coherent and contextually appropriate responses. Embedding models add value across a broad set of NLP use cases such as similarity search, recommendation systems, pattern recognition sentiment analysis and machine translation, as well as computer vision use cases such as image recognition and fault detection.

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Business Impact

Embedding models drive business value by allowing generative AI (GenAI) models to use data that isn't part of their original training to deliver accurate responses. They aid in use cases such as personalization, anomaly detection and search, and support better decision making by leveraging large datasets at scale.

Drivers

- **Data explosion:** With the exponential growth in data generation from various sources like social media, customer transactions and enterprise knowledge management systems, enterprises need efficient ways to process, understand and leverage this vast amount of information. Embedding models help by converting unstructured data into formats that are easier to analyze and act upon by GenAI applications and models, often in conjunction with vector databases.
- **Growing RAG deployments for GenAI use cases:** As companies increasingly invest in RAG architectures and AI agents using RAG, embedding models become essential. They provide the foundational technology for data representation and retrieval in RAG workflows, facilitating more accurate and insightful outcomes.
- **Model as a service:** Embedding models are increasingly offered as a managed API by large AI platform providers, including the large cloud hyperscalers, making them more accessible and easier to manage.
- **Multimodal data integration:** Modern enterprises often have to process and analyze data from diverse sources and formats, including text, images, audio and video. Embedding models facilitate the integration of these multimodal data sources into a unified representation, enhancing data coherence and the quality of insights.
- **Customer experience enhancement:** Enterprises are keen on improving customer engagement and satisfaction. Embedding models support these goals by enabling personalized recommendations, targeted marketing and customer support automation, all of which rely on a deep understanding of customer and industry data.
- **Scalability of AI applications:** As AI applications grow in complexity and size, embedding models offer a scalable solution that can handle increasing data volumes without a corresponding increase in computational demand, making them ideal for deployment in resource-constrained environments.

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Obstacles

- **Data privacy and security concerns:** Handling sensitive information with embedding models — particularly cloud-based models — can introduce security and compliance issues, especially in regulated industries.
- **Data quality and availability:** The quality of embeddings (numerical representation of data) is highly dependent on the quality of enterprise data. Lack of adequate data collection, cleaning and preprocessing can result in suboptimal outcomes from the embedding models.
- **Complex use cases:** When dealing with multimodal documents with structure and connections of concepts across documents, vectorization through chunking may not be enough. More complex metadata management concepts like knowledge graphs are needed.
- **Scalability issues:** Scaling embedding models to handle enterprise-level data volumes and real-time processing needs can be technically challenging.
- **Cost:** Embedding models add an additional layer of cost for the model API, or to the compute and operational overheads of managing the life cycle of the models.
- **Effectiveness:** There could be use cases where using the direct vector representations from an embedding model that has only been generically pretrained may not produce a useful embedding representation of the task. In such scenarios, the embedding models need to be fine-tuned, adding to the costs and complexity and requiring additional know-how.

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User Recommendations

- **Assess data infrastructure:** Ensure the existing data infrastructure can support the demands of embedding models, including data collection, storage, quality and processing capabilities. Consider upgrades if necessary to handle large-scale data efficiently.
- **Evolve enterprise architecture to support GenAI:** Build technology stacks that support embedding models, embedding clustering and knowledge graphs as part of a wider set of tools to deal with the variety of use cases your enterprises need to address.
- **Prioritize data privacy and security:** Address potential security and privacy concerns from the outset by implementing robust data governance policies and ensuring compliance with relevant regulations.
- **Start small:** Begin with pilot projects to demonstrate the value of embedding models. Select use cases that can show quick wins to build support and provide learning experiences.
- **Leverage cloud APIs:** Consider cloud platforms for their scalability and flexibility. Most GenAI cloud platform providers offer embedding models as an API to consume.
- **Measure impact:** Develop metrics to measure the effectiveness and ROI of embedding model initiatives. Use these metrics to justify further investment and guide the scaling of successful applications.

Sample Vendors

AI21 Labs; Amazon Web Services (AWS); Cohere; Google; Meta; Microsoft; Mistral AI; MongoDB (Voyage AI); NVIDIA; OpenAI

Gartner Recommended Reading

How to Improve and Optimize Retrieval-Augmented Generation Systems

Innovation Guide for Generative AI Technologies

Self-Supervised Learning

Analysis By: Pieter den Hamer, Erick Brethenoux

Benefit Rating: Transformational

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Market Penetration: 1% to 5% of target audience

Maturity: Emerging

Definition:

Self-supervised learning (SSL) is an approach to machine learning (ML) in which labels or supervisory signals are created from the data itself, without having to rely on historical outcome data or external (human) supervisors to provide labels or feedback. It is inspired by the way humans learn through observation, gradually building up general knowledge about concepts, events and their relations or spatiotemporal associations in the real world.

Why This Is Important

SSL aims to overcome one of the biggest drawbacks of supervised learning: the need to have access to typically large amounts of labeled training data. SSL can be a solution to the practical problems in many organizations with limited relevant data, or where manual labeling is prohibitively expensive. Moreover, SSL has opened the door to AI use cases where there is an abundance of unlabeled data but a scarcity of labeled data, greatly increasing AI's broadness and versatility.

Business Impact

SSL is best known as the key enabler of large language models (LLMs). It is also the main way in which models are trained for computer vision in contexts like autonomous driving or medical image diagnosis. A growing number of other industries have started experimenting with SSL. For example, for fraud detection in finance, or for quality assurance in manufacturing.

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Drivers

- **Making ML feasible in the absence of labeled training data:** In SSL, labels can be generated automatically from the data itself, without the need for human annotation. This can be done, for example, by masking elements in the available data (e.g., a part of an image, a section of a time series of sensor readings or transactional data points, a frame in a video, or a word in a sentence) and then training a model to "predict" the missing element. For example, the model learns how one part relates to another, how one situation typically precedes or follows another, and which words often go together. As a result, the model increasingly represents the most essential concepts, features, and their spatial, temporal or other relations.

- **Leveraging the internet to create very large foundation models:** SSL (together with new, highly parallelizable deep learning methods) has opened the door to utilizing massive amounts of unlabeled training data. This has enabled the creation of very large multipurpose foundation models such as LLMs.

- **Using both structured and unstructured data:** Most current SSL use cases are based on unstructured, often visual or textual data. However, increasingly, structured data is used as well, particularly where data is unlabeled or needs to be relabeled often, or for more complex or time-series data.

- **Enabling multisource and multimodal learning:** Although each data source (e.g., transactions, documents, news events, social media or sensors) can be valuable in its own right, even more value can be derived by identifying associations between elements and events from multiple data sources. This is true even if they are in different formats or multiple modalities (text, visual, sound, etc.).

- **Stepping toward a next generation of AI:** Advances in SSL are driven by the interest in bringing AI closer to the ways humans learn through associations and incrementally building a world model.

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Obstacles

- **Scarcity of skills and experience:** SSL is currently practiced by a limited number of innovative AI companies. Industries such as healthcare and, to a lesser extent finance, retail and automotive, have begun to apply SSL, but only with low to moderate adoption.
- **Limited tool support:** Although open-source ML frameworks — such as TensorFlow, PyTorch and JAX — support several SSL methods, broader tool support is lacking, making implementation a knowledge-intensive and low-level coding exercise.
- **Risk of bias and evaluation difficulty:** Inherent biases in data can be learned and subsequently amplified. With limited human involvement and no supervised labeling, it's hard to see what's going on and to detect bias or other issues.
- **High costs:** Not only for very large foundation models but also for smaller models, SSL may require significant amounts of training data and compute time. Although such data does not require costly labeling by humans, it still must be vetted and prepared to ensure it's clean, diverse and representative.

User Recommendations

- Use foundation models that are trained via SSL by suppliers. These models may require validation to ensure accuracy or check for bias. They may also need to be fine-tuned using a smaller, possibly synthetic, labeled dataset for supervised learning or by applying reinforcement learning.
- Apply SSL only when its value justifies the risks of a still complex approach. Scarce, highly experienced ML experts are needed to carefully design and implement SSL.
- Employ SSL when manual labeling or data annotation is too expensive or infeasible. Compare SSL with alternative approaches such as data labeling and annotation services, synthetic data, reinforcement learning, active learning, or federated learning.
- Track the developments in SSL. Monitor when more mature SSL can potentially become pervasive for a next generation of applications with AI and ML, not limited to foundation models.

Sample Vendors

aiMotive; craftworks; Google; Meta; Microsoft; SenseTime; Speechmatics; V7; viso.ai; Zeta Alpha

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Artificial General Intelligence

Analysis By: Pieter den Hamer, Philip Walsh

Benefit Rating: Transformational

Market Penetration: Less than 1% of target audience

Maturity: Embryonic

Definition:

Artificial general intelligence (AGI) is the (currently hypothetical) capability of a machine that can match or surpass the capabilities of humans across all cognitive tasks. In addition, AGI will be able to autonomously learn and adapt in pursuit of predetermined or novel goals in a wide range of both physical and virtual environments.

Why This Is Important

With AI's growing sophistication — including the recent advances in generative AI (GenAI) and agentic AI — a growing number of AI experts have shortened their predicted timelines for achieving AGI in the future or view AGI as no longer purely hypothetical. A clear, shared definition of AGI is necessary for evidence-based governance and realistic expectations. Achieving AGI would be a transformative tipping-point with profound consequences for productivity, employment, geopolitical power, legal, ethical and cultural norms — and society at large.

Business Impact

In the near term, anticipation of AGI fuels both overly optimistic expectations and existential fears, skewing investment, distorting trust and accelerating the emergence of new AI regulations. Over the longer horizon, the question of who builds and controls AGI — or other forms of increasingly powerful AI — looms large. Many experts see public stewardship as essential, a prospect that could upend private advantage and redraw entire markets.

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Drivers

- Recent advances and growing interest in multimodal large language models (LLMs), so-called reasoning models and AI agents drive considerable hype about AGI. The massive scaling of deep learning and the availability of huge amounts of data and compute power largely have enabled these advances.
- AI's further evolution toward AGI, as defined here, is increasingly complemented by other partially new approaches, such as knowledge or causal graphs, world models, adaptive AI, embodied AI, composite and neurosymbolic AI, and likely other innovations yet unknown.
- A number of AI vendors are openly discussing and actively researching the field of AGI, creating the impression that AGI lies within reach. However, their definitions of AGI vary greatly and are often open to multiple interpretations. Moreover, other leading AI vendors and experts have dismissed AGI as hype, urging focus on the real impact of AI's growing capabilities.
- Humans' innate desire to set lofty goals is also a major driver for AGI. At one point in history, humans wanted to fly by mimicking bird flight. Today, airplane travel is a reality. The inquisitiveness of the human mind, taking inspiration from nature and from itself, is not going to fizzle out.
- People's tendency to anthropomorphize nonhuman entities also applies to AI-powered machines. The humanlike responses of LLMs and the reasoning-like capabilities of recent AI models have fueled this tendency. Although many philosophers, neuropsychologists and other scientists consider this attribution highly uncertain or going too far, it has created a sense that AGI is within reach or at least is getting closer. In turn, this has triggered massive media attention, several calls for regulation to manage the risks of AGI and a great appetite to invest in AI for economic, societal and geopolitical reasons.

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Obstacles

- Little scientific consensus exists on the meaning of "human intelligence." Any claims about AGI are hard to validate in the face of the enormous complexity of the human brain and mind and such a limited understanding of them.
- Unreliability, lack of transparency and limited abstraction and reasoning of pattern-based capabilities in current AI are not easy to overcome with deep learning's intrinsically probabilistic approach. More data or more compute power for ever-bigger models is unlikely to resolve these issues, let alone to achieve AGI. To realize (and control) AGI will require further technological innovations. Therefore, AGI as defined here is unlikely to emerge in the near future.
- If AGI materializes, autonomous actors likely will emerge that, in time, will be attributed with full self-learning, agency, identity and perhaps even morality. This will open a bevy of considerations about AI's legal rights and trigger profound ethical and even religious discussions. AGI also brings the risk of negative impacts on humans, from job losses to a new, AI-triggered arms race and more. A serious backlash may result, and regulations to ban or control AGI are likely to emerge in the near future.

User Recommendations

- Engage with stakeholders to address excessive optimism or unwarranted pessimism, and create or maintain realistic expectations around AGI. Ground AI strategy in concrete business problems rather than speculative AGI forecasts. Recalibrate the AI portfolio periodically as AI capabilities evolve, while leveraging the complementary strengths of human and artificial intelligence.
- Stay apprised of scientific and innovative breakthroughs that may indicate AGI's possible emergence; however, be aware of the broad range of definitions and views regarding AGI, some strict and some less strict. Meanwhile, keep applying current AI to learn, reap its benefits and develop practices for its responsible use.
- Assess whether AI systems truly meet their specific use-case needs, rather than relying on generic measures of intelligence.
- Prepare for emerging AI regulations and promote internal AI governance to manage current and emerging AI risks. Because although AGI as defined here is not a reality now, current AI already poses significant risks regarding ethics, reliability and other areas.

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Sample Vendors

Aigo; Amazon; Anthropic; Butterfly Effect; DeepSeek; Google; Microsoft; OpenAI

Gartner Recommended Reading

Artificial General Intelligence: 5 Future Perspectives and Present Imperatives

Video: Why Artificial General Intelligence (AGI) Matters — and Why It Doesn't

How to Think About Artificial General Intelligence

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At the Peak

AI Agent Development Frameworks

Analysis By: Tigran Egiazarov, Arun Chandrasekaran, Gary Olliffe

Benefit Rating: Transformational

Market Penetration: 5% to 20% of target audience

Maturity: Emerging

Definition:

AI agent development frameworks accelerate the creation of AI agents and AI-powered applications that can autonomously perform complex tasks. These frameworks offer high-level abstractions for large language model (LLM)-based orchestration, along with low-level control over agent logic, memory, planning and API access. They evolved from LLM orchestration libraries into more sophisticated frameworks supporting multiagent modularity, long-term memory and common reasoning patterns.

Why This Is Important

AI agent development frameworks reduce the complexity of building AI agents and AI-powered applications and allow developers to focus on functionality and architecture by abstracting the access to LLMs, memory management, agent orchestration and tools. Agents can automate processes by interpreting and reacting to evolving business inputs without rewriting code, relying instead on updates to business context or process structure. This enables rapid iteration, increased agility, and automation of low- to mid-complexity tasks with less engineering overhead.

Business Impact

AI agent development frameworks enable existing static applications to transform into intelligent applications capable of contextual reasoning and dynamic task execution. With growing demand for agents capable of autonomous planning, multitool interaction (such as model context protocol) and learning, these frameworks play a foundational role in operationalizing AI applications and agents to support business and customer needs.

Drivers

- Significant interest in LLM-based AI agents fuels the need for low-entry barriers in building AI agents, resulting in the growth of AI agent development frameworks.

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- Advances in the capabilities of AI large language models and large reasoning models are enhancing the perception, reasoning and planning abilities of AI agents to support increasingly complex business use cases.
- Real-world use cases are emerging: document processing, research agents, content generation, interaction with GUI like computer-using agent mode, and even code writing assistants using AI agents.
- The frameworks abstract model handling, allowing developers compose, deploy and iterate on agents using cloud-based LLMs, local models or hybrid backends.
- Open-source communities are significantly contributing to the development of AI agent frameworks, resulting in more feature-rich solutions.

Obstacles

- Most frameworks remain below version 1.0 and are evolving rapidly, which may introduce breaking changes.
- While powerful, many tools lack built-in guardrails, requiring developers to manually implement evaluation, fallback and safety mechanisms.
- Most frameworks are Python-centric, making integration with enterprise applications in other languages more complex.
- The nondeterministic nature of AI agents does not allow them to fully replace the “standard” applications and core systems due to the risks of an error. This therefore slows adoption of AI agents.
- Multiagent workflows are harder to manage at scale and require thoughtful orchestration and observability design. They also increase inference costs, requiring judicious selection of foundation models and iteration settings.

User Recommendations

- Start with well-scoped, low-risk use cases such as document summarization or back-office customer support.
- Identify the use cases where you require the highest level of customization for AI agents beyond what typical AI agent no-code builders or robotic process automation tools can offer.

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- Select the frameworks based on the several criteria such as multiagent support, access to ecosystem and tools coverage, memory support, ease of use, and code execution.
- Use cloud-native compute models (e.g., function as a service, containers) for scalable deployments. Integrate observability tools (e.g., LangSmith, MLflow) for debugging and tracing. Constantly evaluate AI agents with eval tools and frameworks (e.g., Arize AX, Confident AI DeepEval) while setting guardrails around the AI agents.
- Invest in compatible architecture and an API-first approach, RAG pipelines and data governance to support flexible agent behavior.
- Be aware that building complex, distributed AI-agentic systems requires significant investment in API-driven orchestration, as nondeterministic workflows can impact performance.
- Follow open-source communities closely for rapid improvements and emerging best practices.
- Tackle the coordination, complexity and communication challenges in AI agent design (see Emerging Patterns for Building LLM-Based AI Agents).

Sample Vendors

Akka; Alibaba; Baidu; CrewAI; Google; Hugging Face; LangChain; Meta; Microsoft; OpenAI

Gartner Recommended Reading

Innovation Insight: AI Agents

Use AI-Ready Data With Composable Architecture to Deliver Intelligent Applications

Innovate Business Models Using AI Agents

How to Price Your AI Agents

Domain-Specific GenAI Models

Analysis By: Jim Hare, Leinar Ramos, Eric Goodness, Roberta Cozza

Benefit Rating: High

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Market Penetration: 1% to 5% of target audience

Maturity: Adolescent

Definition:

Domain-specific generative AI (GenAI) models are tailored for the needs of specific industries, business functions or workflows. They enhance the accuracy, privacy and compliance while minimizing hallucinations and prompt engineering. Built from scratch or fine-tuned on domain data, these models outperform general-purpose models targeted in domains like healthcare, finance and law.

Why This Is Important

Although general-purpose models perform well across a broad set of applications, they may be impractical for many use cases that require domain-specific data. Domain-specific GenAI models can improve use-case alignment within the enterprise while delivering improved accuracy, security and privacy, and better contextualized answers, reducing the need for advanced prompt engineering. Through targeted training, these models can lower hallucination risks associated with large models.

Business Impact

Domain-specific GenAI models facilitate:

- Faster time to value for AI projects by providing a more advanced starting point for industry-specific tasks.
- Improved performance, compliance and trust by reducing inaccuracies and hallucinations using techniques such as model training or fine-tuning on domain-specific data sources.
- Broader adoption and improved user trust of GenAI because outputs align with domain logic and/or roles, helping organizations apply them to use cases where general-purpose models are not performant enough.

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Drivers

- **Proliferation of open generative models:** Open foundation models are emerging as a viable source for domain-specific models, rapidly closing the performance and reliability gap with proprietary models, and offering a cost-effective and flexible alternative for model training and specialization.
- **Increased specialization in GenAI model use:** As GenAI adoption in organizations moves from general use cases to more specific ones, targeted models that have been trained or fine-tuned with industry-specific data for domain-specific use cases will become more important.
- **Expanded vendor domain offerings:** Demand is increasing for GenAI in many industries, such as healthcare, life sciences, legal, financial services and the public sector. Technology service providers are looking to meet this increased demand by offering products and services tailored to the needs of these industries, including domain-specific models.
- **Limitations of general-purpose models:** General-purpose models often require significant prompt engineering or fine-tuning along with security guardrails to optimize the output for enterprise use cases because the models haven't been trained using industry or business function data. Domain-specific models can be more easily adapted to specific requirements, increase context, utilize available resources more efficiently and deliver powerful language processing capabilities that address specific challenges and tasks.
- **Growing interest in domain-specific AI agents:** The current hype around agentic AI is fueling rapid growth in domain-specific AI models. Agentic AI are autonomous or semiautonomous software entities that use AI techniques to perceive, make decisions, take actions and achieve goals in their digital or physical environments. Enterprises are realizing the value of specialized agents that deeply understand and operate within constrained contexts like legal, medical or engineering domains. Therefore, there is a strong push to use tailored models optimized for those verticals, combining general AI capabilities with domain-specific expertise.

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Obstacles

- **Data scarcity and quality:** Domain-specific models rely heavily on high-quality, domain-relevant data, which is often limited in volume, harder to obtain and of low quality, requiring data clean-up.
- **Reduced model versatility:** By optimizing for a narrow set of use cases, domain-specific models sacrifice versatility, losing the ability to perform well across a wider set of tasks and domains when compared with general-purpose models.
- **Model proliferation:** An organization might need different domain-specific models for different use cases, and thus have an increased portfolio of models to manage.
- **Inaccuracies and hallucinations:** Even though inaccuracies might reduce in the target domain compared with general-purpose models, domain-specific models can still make mistakes and hallucinate.
- **Maintenance and updating:** Once deployed, domain-specific models may require updating because of fast-changing knowledge bases, resulting in stale models, especially in fields like medicine, finance or cybersecurity.

User Recommendations

- Consider domain-specific models for situations that require accurate and contextually appropriate outputs within a specific industry or business function. While domain-specific models offer tailored capabilities, they still require retrieval-augmented generation to effectively incorporate and leverage an enterprise's proprietary knowledge base.
- Look for off-the-shelf domain-specific models for your specific use case before deciding to build your own custom domain model. The development of custom domain models poses resource-related issues, predominantly in terms of computational resources and specialized skills.
- Ask domain-specific model providers about the specific use cases for their model, along with the pedigree, amount and quality of the data used to train the model.
- Establish life cycle management practices to avoid model degradation and ensure continuous compliance with regulatory or industry standards.

Sample Vendors

Arcee AI; Bloomberg; ChatClimate; Google; Harvey; Kasisto; LeewayHertz; Mistral AI; Spellbook; Vic.ai

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Gartner Recommended Reading

2024 Tech Provider Top Trends: Vertical Generative AI Models

TSP 2025 Trends: Domain-Specific Models Will Displace Generic LLMs in the Enterprise

Prompt Engineering

Analysis By: Frances Karamouzis, Jim Hare

Benefit Rating: High

Market Penetration: 5% to 20% of target audience

Maturity: Emerging

Definition:

Prompt engineering is the discipline of providing inputs, in the form of text or images, to GenAI models to specify and confine the set of responses the model can produce. The inputs prompt a set that produces a desired outcome without updating the actual weights of the model (as done with fine-tuning). Prompt engineering is also referred to as "in-context learning," where examples are provided to further guide the model.

Why This Is Important

Prompt engineering is the linchpin to business alignment for desired outcomes. It is important because LLMs and GenAI models in general are extremely sensitive to nuances and small variations in input. A slight tweak can change an incorrect answer to one that is usable as an output. Each model has its own sensitivity level, and the discipline of prompt engineering is to uncover the sensitivity through iterative testing and evaluation.

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Business Impact

- **Well-structured prompts:** Increase the quality of responses
- **Performance:** Helps improve model performance and reduce hallucinations
- **Business alignment:** Allows various roles to steer foundation models, which are general-purpose in nature, to align to the business, domain and industry
- **Time to market, quality, efficiency and effectiveness:** There are a number of architecture and execution options that various users must balance. There are also many prompt optimization tools that will diminish (or at least shift) the need for manual engineering

Drivers

- **Balance and efficiency:** The fundamental driver for prompt engineering is it allows organizations to strike a balance between consuming an “as-is” offering, versus pursuing a more expensive and time-consuming approach of fine-tuning. GenAI models, and in particular LLMs, are pretrained, so the data that enterprises want to use with these models cannot be added to the training set. Instead, prompts can be used to feed content to the model with an instruction to carry out a function.
- **Process or task-specific customizations or new use cases:** The insertion of context and patterns a model uses to influence the output generated allows for customizations for a particular enterprise or domain, or for regulatory items. Prompts are created to help improve the quality for different use cases — such as domain-specific question answering, summarization and categorization — with or without the need for fine-tuning a model, which can be expensive or impractical. This would also apply to creating and designing new use cases that utilize the model’s capability for image and text generation.
- **Validation and verification:** It is important to test, understand and document the limits and weaknesses of the models to ensure a reduced risk of hallucination and unwanted outputs.

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Obstacles

- **Approaches and techniques:** A unified approach to performing prompt engineering does not exist. Complex scenarios need to be broken down into smaller elements. It is challenging to debug complex prompts. Understanding how specific prompt elements influence the logic of the LLM is vital.
- **Dynamic vendor (provider) landscape:** Providers are constantly fine-tuning and tweaking techniques and approaches that influence behavior within their offerings. Organizational teams creating prompts must iteratively adapt and pivot to ensure scalable and maintainable methods of prompt engineering.
- **Scalability:** Scaling prompt engineering for large projects or across multiple teams can be difficult due to the manual and iterative nature of the process. Scalable and maintainable methods of prompt engineering are still a work in progress for most organizations.

User Recommendations

- Recognize and embrace the underlying premise that strong prompt engineering is directly correlated to quality results and outputs. And this impacts cycle time and speed to outcome.
- Invest in the skill of prompt engineering as a “must-have” for a variety of roles in the organization; for example, data scientists, business users, domain experts, software engineers and citizen developers.
- Elevate the team with the myriad of options of prompt optimization tools that will diminish (or at the very least shift) the need for manual engineering.
- Communicate and cascade the message that prompt engineering is not foolproof. Enterprise teams apply rigor and diligence to permeate and work to ensure successful solutions.

Sample Vendors

FlowGPT; Google; HoneyHive; Microsoft; PromptBase; Salesforce

Gartner Recommended Reading

How to Engineer Effective Prompts for Large Language Models

Prompt Engineering With Enterprise Information for LLMs and GenAI

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Quick Answer: How Will Prompt Engineering Impact the Work of Data Scientists?

Generative AI Changes Software Engineering Leaders' Responsibilities

AI TRiSM

Analysis By: Avivah Litan, Jeremy D'Hoinne, Bart Willemsen, Lauren Kornutick, Max Goss, Andrew Bales, Sumit Agarwal

Benefit Rating: High

Market Penetration: 5% to 20% of target audience

Maturity: Early mainstream

Definition:

AI trust, risk and security management (TRiSM) comprises four layers of technical capabilities that support enterprise policies for all AI use cases and help assure AI governance, trustworthiness, fairness, safety, reliability, security, privacy and data protection. The top two layers — AI governance, and AI runtime inspection and enforcement — are new to AI and are, in part, consolidating into a distinct market segment. The bottom two layers represent traditional technology focused on AI.

Why This Is Important

AI brings new trust, risk and security management challenges that conventional controls do not address. Enterprises are most concerned with data compromise, third-party risks, and inaccurate or unwanted outputs, and need to ensure enterprise AI behavior and actions align with enterprise intent. Enterprises must further retain independence from any single AI model or hosting provider to ensure scalability, flexibility, cost control and trust, as AI markets rapidly mature and change.

Business Impact

Organizations that do not consistently manage AI risks are exponentially inclined to experience adverse outcomes such as project failures, AI misperformance and compromised data confidentiality. Inaccurate, unethical or unintended AI outcomes, process errors, uncontrolled biases, and interference from benign or malicious actors can result in security failures, financial and reputational loss, or liability and social harm. AI misperformance can also lead organizations to make suboptimal or incorrect business decisions.

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Drivers

- The increasing use of AI and GenAI is limited by a lack of trust in AI as a safe and ethical option for supporting critical business processes.
- Enterprises face multiple AI risks and are most concerned with data compromise, third-party risks, and inaccurate or unwanted outputs.
- Malicious hacks against enterprise AI are still uncommon, while incidents of unconstrained harmful chatbots are well-documented, and internal oversharing data compromises are prevalent.
- User demand for GenAI TRiSM solutions is steadily increasing, and providers of all sizes are competing for this new enterprise business.
- Some organizations are mostly concerned with security and risk mitigation, while others also focus on supporting ethical or safe practices and regulatory compliance.
- AI trust, risk and security issues surface organizational silo issues, pushing teams to realign to solve problems that cross departmental boundaries and to implement technical measures that address them.
- The introduction of LLM-based AI agents with differing degrees of agency allows AI to take actions with or without human intervention.
- The rapid proliferation of AI agents will create more need for governance than human-in-the-loop oversight can fulfill alone.
- Regulations for AI risk management are driving businesses to institute measures for managing AI risk. Such regulations define new compliance requirements that organizations will have to meet on top of existing ones, like those pertaining to privacy protection.

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Obstacles

- Adopting AI TRiSM technology is often an afterthought. Organizations generally don't consider it until AI applications are in production, when it becomes challenging to retrofit.
- Many enterprises are resource-constrained and don't have the skills or capacity to implement TRiSM technology.
- Enterprises often rely on their incumbent vendors to provide AI functionality for TRiSM capabilities though they often lack it, and must rely on vendor licensing agreements to ensure their confidential data remains private in the host environment.
- Off-the-shelf software that embeds AI is often closed and does not support APIs to third-party AI TRiSM products that can enforce enterprise policies.
- Most AI threats and risks are not fully understood and not effectively addressed.
- AI TRiSM requires a cross-functional team, including legal, compliance, security, IT and data analytics staff, to establish common goals and use common frameworks.
- While AI TRiSM can integrate life cycle controls, this poses significant implementation challenges especially with embedded AI that often lacks API support.

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User Recommendations

- Set up an organizational unit to manage AI TRiSM that is integrated with the organization's overarching governance initiative. Include members with a vested interest in AI projects.
- Define acceptable use policies at a level granular enough to enforce.
- Discover and inventory all AI used in the organization, leveraging the capabilities of AI TRiSM vendors who support this.
- Revisit and implement data classification, protection and access management across all enterprise information — both structured and unstructured — that can potentially be utilized by AI. Collaborate across different teams involved in information governance.
- Work with legal and compliance to contractually define accountability and responsibility for unacceptable AI use or behavior in third-party-embedded AI applications.
- Obtain vendor attestation to meet legal requirements.
- Evaluate and implement layered AI TRiSM technology to continuously support and enforce policies across all AI entities in use. This includes enterprise-owned AI TRiSM services, along with controls offered by frontier model providers, but do not solely rely on the latter.

Gartner Recommended Reading

Market Guide for AI Trust, Risk and Security Management

Top Strategic Technology Trends for 2024: AI Trust, Risk and Security Management

The Cybersecurity Leader's Playbook: Navigating the EU AI Act

GenAI Application Orchestration Frameworks

Analysis By: Arun Chandrasekaran, Tigran Egiazarov

Benefit Rating: High

Market Penetration: 5% to 20% of target audience

Maturity: Adolescent

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Definition:

Generative AI (GenAI) application orchestration frameworks provide an abstraction layer to enable prompt chaining, model chaining, interfacing with external APIs, retrieving contextual data from data sources and maintaining statefulness (or memory) across various model requests. These frameworks also provide templates, monitoring, evaluation and deployment capabilities for new GenAI applications and AI agents.

Why This Is Important

GenAI application orchestration frameworks' functionalities expand AI foundation models' capabilities, making them more adaptable, interactive, context-aware and efficient in various applications. GenAI application orchestration frameworks enable seamless application workflows by providing a standard interface to GenAI models. These frameworks enable data integration, chaining prompts and models, effective prompting through prompt templates, input prompt optimization and output parsing. The frameworks are expanding now into agent development, orchestration, governance and observability.

Business Impact

GenAI application orchestration frameworks enable production-ready, composable and extensible AI applications and agents. These frameworks allow teams to build, iterate and deploy GenAI apps quickly, reducing time from idea to production, enabling faster time to value. They also enable businesses to lower vendor lock-in and reduce cost, through usage of multiple models, often smaller, cheaper models, for low-complexity tasks and reserve larger models for high-value tasks.

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Drivers

- **Use-case evolution:** Most AI applications require composite AI capabilities or an ensemble of AI models, which unify a number of different models that are combined to enable the right balance of features to enable application quality, cost, performance and risk. This typically requires complex, multicomponent AI systems. Models may need to be chained together, routed between or augmented with data retrieval or evaluation or guardrail steps. This drives the need for orchestration frameworks that support complex, production-ready AI applications.
- **GenAI application development:** Developers need to capitalize on GenAI models to build applications. GenAI application orchestration frameworks provide developers with a new way to build user interfaces (UIs) and automate application builds.
- **Need for GenAI customization:** GenAI models can be combined with enterprise data through advanced prompt engineering techniques or model fine-tuning. GenAI application orchestration frameworks provide an open approach to integrating data sources with AI models.
- **Growing interest in AI agents:** Although AI foundation models are a great advance, they still require extensive human intervention (in terms of prompts) to achieve business outcomes. GenAI application orchestration frameworks enable developers to build agents to reason about problems and break them into smaller, executable subtasks. AI agents dynamically interact with their environments to achieve goals. Orchestration tools can enable AI agents to be stateful, integrate with a wide array of data sources and APIs for knowledge and decision making and automatically orchestrate workflows for task completion.

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Obstacles

- **Immaturity:** Although the tools are gaining developer mind share, there are still gaps in meeting enterprise needs for stability and robust integration. Therefore, enterprise AI teams must cautiously navigate software bugs and security issues, characteristic of most early-stage projects.
- **Lack of clear winners:** Although these orchestration frameworks are designed to bridge GenAI application development, their longevity and ability to innovate iteratively remain unknown.
- **Integration challenges:** Most enterprises operate with a mix of legacy systems and modern applications, making integration with GenAI orchestration frameworks difficult.
- **Lack of awareness:** These tools are new and there is a lack of understanding of what they do, which one to use and how to safely deploy them.
- **Customization and flexibility:** Although these tools offer general capabilities for integrating GenAI models into application workflows, specific enterprise needs might require more customization than is currently available.

User Recommendations

- Encourage experimentation on what these tools do and their potential fit with your technical architecture to reduce vendor lock-in and enable better integration.
- Identify the use cases in which you are implementing prompt engineering, data retrieval or fine-tuning, which can benefit from these tools.
- Select tools that offer APIs and customization options that align with your current and future needs.
- Explore the autonomous agent capabilities of these tools cautiously, given the black-box nature of machine-to-machine interaction. Seek an adequate balance between capabilities, integration, governance and cost-efficiency.
- Take a centralized platform approach to achieve standardization and automation across the GenAI applications you are building.

Sample Vendors

Amazon Web Services (AWS); deepset; Google; Hugging Face; LangChain; LlamaIndex; Microsoft; OpenAI

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Gartner Recommended Reading

Innovation Guide for Generative AI Models

A CTO's Guide to the Generative AI Technology Landscape

Multimodal Generative AI

Analysis By: Danielle Casey, Roberta Cozza

Benefit Rating: Transformational

Market Penetration: 5% to 20% of target audience

Maturity: Emerging

Definition:

Multimodal generative AI is the ability to combine multiple types of data inputs and outputs in generative models, such as images, videos, audio, text and numerical data. Multimodality augments the usability of AI by allowing models to interact with and create outputs across various modalities.

Why This Is Important

Multimodal GenAI is important because real-world data is multimodal. Multimodality helps capture the relationships between different data streams and scales the generation and automation benefits of GenAI across applications and workflows. Multimodal functionality will increasingly be present in tech offerings as foundation models lower the technical barrier to multimodal functionality and as users demand associated performance.

Business Impact

Multimodal GenAI is increasingly having a transformational impact on enterprise applications by enabling the addition of new features and functionality otherwise unachievable. The impact of multimodality is not limited to specific industries or use cases, and can be applied at any touchpoint between AI and humans. Today, text, voice and image are common modalities. Over the next few years, models will evolve to support additional modalities that are more emerging, domain-specific or complex.

Drivers

Multimodal GenAI is being driven by:

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- **Model innovation:** GenAI models are lowering the technology barrier for providers to offer multimodal functionality in their offerings. The emergence of domain-specialized models will also drive multimodal innovation. Models trained on industry-, role- or use-case-specific data will be increasingly multimodal (e.g., legal, finance, healthcare, marketing and multimedia).
- **Improved user experience (UX):** Multimodality improves UX by enabling richer experiences as it meets users with the data that is available. The ability to interact with both unstructured and structured data is supporting additional use cases within the organization.
- **Extensibility of automation:** Multimodality supports new tasks, such as data extraction, data manipulation (converting one data type to another) and creating new data outcomes. Multimodal applications will have a higher automation potential, particularly with the support of domain-specific data in vertical environments and business roles. By 2030, multimodality will increase AI automation potential by 50% with new functionality and use cases.

Obstacles

Multimodal GenAI models are inhibited by:

- **Training challenges:** Multimodal GenAI models use deep learning, data alignment and fusion techniques to train, integrate and analyze multimodal data. Multimodal data has varying degrees of quality and formats compared to unimodal data, amplifying challenges with training cost and time, output accuracy and de-bias.
- **Lack of data:** Data availability may be limited in some modalities (e.g., large-scale audio datasets or healthcare imagery), which impacts model training and accuracy.
- **Data governance:** Multimodal GenAI increases the exposure to a wider range of sensitive data. Examples of particularly sensitive data types include maps or geolocation data, biometric data or health data. Also, regulations and standards are a work in progress and lagging GenAI's capability advancement.
- **Lack of application experience:** Best practices, testing methods, software libraries and supporting tools have not been fully developed to enable multimodal applications.

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User Recommendations

- Identify two or three modalities that are most important to your organization when choosing which GenAI model to use.
- Evaluate and explore use cases where multimodal capabilities may offer differentiating capabilities.
- Assess the technical complexities of processing and integrating data inputs and outputs from diverse multimodal sources. Then, validate early on how these can best be integrated with key legacy or more current workflows.
- Create specific cross-modal data policies and tools aimed at protecting privacy, detecting bias and ensuring compliance to emerging AI regulations.

Sample Vendors

Alibaba Cloud; Anthropic; Archetype AI; DeepSeek; Google; Microsoft; Mistral AI; OpenAI; Stability AI; TwelveLabs

Gartner Recommended Reading

Emerging Tech: Multimodal Generative AI Interfaces Transform User Experiences

Emerging Tech Impact Radar: Generative AI

Emerging Tech: Data Fabrics With Multimodal Data Focus for Generative AI-Enabled Applications

Emerging Tech: Unlock Multimodal GenAI's Potential to Drive Differentiation

Agentic AI

Analysis By: Erick Brethenoux, Pieter den Hamer

Benefit Rating: Transformational

Market Penetration: 1% to 5% of target audience

Maturity: Emerging

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Definition:

Agentic AI is an approach to building AI solutions based on the use of one or multiple software entities that are classified, completely or at least partially, as AI agents. AI agents are autonomous or semiautonomous software entities that use AI techniques to perceive, make decisions, take actions and achieve goals in their digital or physical environments.

Why This Is Important

Agentic AI is at the peak of the Hype Cycle. The agentic AI trend has seen a rapid surge of interest, with AI-agent-related inquiries increasing by 750% in 2024. For no reasons other than confusion entertained by the market, agentic AI is riding the generative AI hype wave. For the appropriate use cases, agentic AI can provide remarkable value with better AI integration. AI agents usher an era of radically different software practices, leading to highly distributed decision systems.

Business Impact

As original AI agent systems have shown for decades, often based on embedded systems, agentic AI can generate significant business value, when appropriately applied. Agentic AI holds the promise to create that value through goal-driven systems by bringing more flexibility, adaptability and higher levels of not only automation but, more importantly, augmentation, further bridging the gap between humans and machines.

Drivers

- **Technological advancement:** Agentic AI incorporates advancement in composite AI (aka hybrid AI), decision intelligence and large action model evolutions; most are quite rapid these days.
- **Market momentum and hype:** Agentic AI benefits from a formidable market momentum, generating significant investments from many user organizations currently experimenting with the technology.
- **Vendor investment:** Many vendors have accelerated capabilities for agentic AI, contributing to the acceleration of the trend and the general demand.
- **The advanced automation promise:** Agentic AI promises less brittle, more resilient and contextual process flows compared to conventional methods like workflow automation and RPA, opening doors to flexible levels of automation.
- **Complex use cases:** The increasing complexity and dynamics in environments, goals or execution within business contexts are indicative of a need for more distributed and less deterministic characteristics, which multiagent systems can provide.

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Obstacles

- Market hype is diluting the meaning of AI agents, with vendors engaging in “agent washing” — rebranding existing products, such as AI assistants, RPA tools and chatbots, to capture buyers’ attention, without offering substantial agentic capabilities. This creates false expectations about the maturity of the technology.
- Predictability is a core limitation for AI agents. Current AI agents may be capable of performing certain tasks, but they are unable to do these reliably enough for a given use case. This makes most of them unsuitable for full automation.
- The increased autonomy of AI agents means they present a series of new risks above and beyond the ones that come with using stand-alone AI models and GenAI assistants.
- Most products in the market are AI assistants, not AI agents. AI assistants do not typically engage in self-directed actions like AI agents do. Yet users are getting emboldened, not realizing the full commitments required to build agents.

User Recommendations

- Do not take your services or software providers’ promises for granted. Assess the capabilities to determine whether their products truly qualify as AI agents or agentic AI (and to what extent), and can actually deliver the expected benefits. Chances are that in many cases, you will not need those capabilities.
- Determine the extent to which AI agent capabilities are needed by applying criteria regarding the complexity, dynamics and other relevant characteristics of each use case.
- Evaluate and keep an open mind about alternative delivery approaches — such as conventional workflow automation or robotic process automation — to avoid using AI agents in use cases where they are not the best solution.
- Prepare for the future of AI agents by laying the foundations of application composability, governance and data that will be required.

Sample Vendors

CrewAI; Dust; Epic Games; LangChain; Unity; XMPro

Open-Source LLMs

Analysis By: Eric Goodness

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Benefit Rating: High

Market Penetration: 20% to 50% of target audience

Maturity: Adolescent

Definition:

Open-source large language models (LLMs) are deep-learning foundation models distinguished by their terms of use, distribution granted to developers, and developers' access to the source code and model architecture. Open-source LLMs are made available to the public through a license that enables anyone to access, use, modify and distribute the model source code without restriction.

Why This Is Important

Open-source LLMs, frameworks, libraries, tools and datasets accelerate enterprise value from generative AI (GenAI) implementation by democratizing commercial access and allowing developers to optimize models for specific tasks and use cases. Additionally, open-source LLMs provide access to developer communities in enterprises, academia and other research roles that are working toward common goals to improve and make the models more valuable.

Business Impact

Open-source LLMs increase innovation potential through customization, better control over privacy and security, the ability to leverage collaborative development, model transparency and the potential to reduce vendor lock-in. Ultimately, open-source LLMs offer enterprises smaller models that are easier and less costly to train, and enable business applications and core business processes.

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Drivers

- The increasing interest in fine-tuning LLMs is a key driver of open-source adoption. Open-source LLMs offer the flexibility of customization due to accessible model parameters and source code, allowing developers to tailor models and control costs, output and alignment. Furthermore, basing products on open-source LLMs enables continuous development grounded on internal and customer needs, making applications harder for competitors to imitate.
- Active, communally driven innovation and support fuel the interest in and adoption of open-source LLMs. These models benefit from the collaborative efforts of development communities that continuously refine them. Additionally, some communities fine-tune these models for specific uses. The need for transparency also favors open-source LLMs, as proprietary LLMs lack transparency in architectures, training, datasets and model weights. Open-source options allow for developer inspection and analysis.
- A growing need to avoid vendor lock-in for LLMs exists, even with commercial support. Given the rapid evolution of the GenAI landscape, open-source LLMs provide the flexibility to switch models or providers more easily.
- Interest in small language models (edge LLMs) is rising due to their ability to be trained and fine-tuned for specific domains at reduced costs. Customizing closed-source models for this purpose is significantly more expensive and complex.

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Obstacles

- Investments in personnel, tooling and infrastructure to train and run open-source LLMs can be high. This represents a significant fixed cost, compared with proprietary alternatives, as well as longer time to value and an impediment to implementation.
- Upgrading and managing open-source model life cycles will be difficult, particularly if a lot of customization is built on top of a given version.
- The complexity of licensing models in open source impedes implementation. Open source may impose restrictions on the consumer, requiring rigorous review from legal teams before adoption as not all models are certified for commercial use. Many users fear exposure to legal liability if an open-source model is trained on copyrighted materials. While many open-source models publish the provenance of training data, some do not.
- As measured by various benchmarks, the gap in accuracy between proprietary LLMs and open-source LLMs reduces demand. This gap becomes less of a factor for smaller targeted LLMs.

User Recommendations

- Perform due diligence to understand the legal exposure related to training data, and the potential biases in the models, such as verifying the open-source license and checking for restrictions on its commercial use.
- Investigate to ensure that data privacy and security measures are in place when using open-source LLMs to process sensitive information.
- Engage proactively with open-source LLM communities as part of the due diligence process to discern the positives and drawbacks associated with various models.
- Evaluate different open-source LLMs, based on factors such as performance, resource requirements, compatibility and documentation. Test the models on sample data, and compare their outputs against your defined objectives.
- Consider experimenting with fine-tuning on domain-specific data to assess the LLM's adaptability to your specific use cases.

Sample Vendors

AI21 Labs; Alibaba Cloud; Cerebras; DeepSeek; Google; H2O.ai; Meta; NVIDIA; Stability AI

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Gartner Recommended Reading

Answering IT Leaders' Top 10 Questions on Open Generative AI Models

Quick Answer: How Do I Compare and Evaluate LLMs?

How to Create and Enforce a Governance Policy for Open-Source Software

Hype Cycle for Open-Source Software, 2024

Market Guide for AI Trust, Risk and Security Management

GraphRAG

Analysis By: Afraz Jaffri, Yogesh Bhatt

Benefit Rating: Moderate

Market Penetration: 5% to 20% of target audience

Maturity: Adolescent

Definition:

GraphRAG is a technique to improve the accuracy, reliability and explainability of retrieval-augmented generation (RAG) systems. The approach uses knowledge graphs (KGs) to improve the recall and precision of retrieval, either directly by pulling facts from a KG or indirectly by optimizing other retrieval methods. The added context refines the search space of results, eliminating irrelevant information.

Why This Is Important

In an enterprise context, RAG techniques have problems of veracity and completeness of responses due to the limited accuracy of retrieval, contextual understanding and response coherence. KGs, a well-established technology, can represent data within documents and other data sources, and the associated metadata. Combining both aspects allows RAG applications to retrieve text based on the similarity to the question and contextual representation of the query and corpus, improving response accuracy.

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Business Impact

Enterprise generative AI applications based on the RAG architecture can benefit from decreased time to value. Currently, moving applications from 70% to 80% accuracy onwards requires multiple iterations and tuning of parameters. KGs can decrease the number of iteration cycles. They are key to supporting AI-ready data. Organizing data in a KG has added benefits beyond RAG implementation for metadata management, catalogs, accessibility and semantic representation.

Drivers

- Standard RAG approaches fail to meet the needs of many enterprise use cases where high thresholds of accuracy need to be guaranteed. GraphRAG can increase accuracy rates by representing context, rules and information in a single queryable model.
- KGs are increasingly relevant for data management, semantic enrichment, knowledge representation and data integration.
- An increasing focus on the combination of symbolic and nonsymbolic AI models to achieve higher levels of comprehension, reasoning and explainability have driven the adoption of GraphRAG.
- The addition of vector search capabilities inside KG platforms and databases eases implementation difficulties of integrating separate vector and KG retrieval.
- Graph databases are already an established part of many enterprises' data landscape, providing a natural starting point for GraphRAG use-case implementation.
- The flexibility of graph data models provides a method for answering questions that can be retrieved by structured queries, reducing the need for large language models (LLMs) with high token input and output costs.

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Obstacles

- The creation of KGs for use in GraphRAG architectures requires a high level of expertise in data and domain knowledge modeling. LLMs are helping to augment the process but still need to be guided by a domain expert.
- GraphRAG architectures have not been tested at scale for performance and scalability.
- The rapidly growing number of RAG approaches is causing confusion and indecision on which method or approach to use for long-term viability.
- The evaluation of GraphRAG, as with other RAG approaches, is difficult and requires the creation of custom metrics.
- Using a GraphRAG approach requires high-quality data and additional metadata, which may not be present in sufficient quantities for solutions that require immediate delivery.

User Recommendations

- Include the GraphRAG approach in building RAG systems by creating services that can easily transform data into KGs using associated metadata.
- Utilize the GraphRAG approach for nontrivial RAG use cases or for those where error tolerance levels are low.
- Use domain experts to verify and enhance the construction of KGs for use in GraphRAG systems by creating semantic layers that include domain-specific terminology.
- Expand existing methods for metadata extraction and management to include KG models that provide context for downstream applications.

Sample Vendors

Altair; Amazon Web Services (AWS); ArangoDB; Franz; Graphwise; Microsoft; Neo4j; Ontotext; Squirro; Writer

Gartner Recommended Reading

How Large Language Models and Knowledge Graphs Can Transform Enterprise Search

How to Evaluate the Applicability of Knowledge Graphs for Your Use Cases

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How to Build Knowledge Graphs That Enable AI-Driven Enterprise Applications

How to Improve and Optimize Retrieval-Augmented Generation Systems

AI-Augmented Software Engineering

Analysis By: Arun Batchu, Manjunath Bhat, Nitish Tyagi, Keith Holloway

Benefit Rating: Transformational

Market Penetration: 5% to 20% of target audience

Maturity: Adolescent

Definition:

AI-augmented software engineering (AIASE) refers to the integration of AI as a collaborative partner to enhance the capabilities of software engineers across the software development life cycle (SDLC). It aims to accelerate the delivery of high-quality software by leveraging AI technologies in various phases, including planning, design, development, testing, validation, security, deployment and maintaining applications.

Why This Is Important

The SDLC encompasses routine tasks within both the creative and operational DevSecOps loops. AI automation reduces manual effort, allowing engineers to focus on innovation, reduce technical debt, enhance quality and security, improve team collaboration, and lower operational costs. AI technologies augment engineers' cognitive tasks, including analyzing logs, optimizing configurations and generating scripts, code, unit tests and documentation.

Business Impact

AIASE transforms the SDLC by integrating AI as a collaborative partner. This integration accelerates the delivery of high-quality software and enhances various phases of development, including planning, design, testing and maintenance. By automating routine tasks and augmenting cognitive tasks such as log analysis and code generation, AIASE reduces manual effort, allowing engineers to focus on innovation. This leads to reduced technical debt, improved quality and security, enhanced team collaboration and lower operational costs, ultimately providing transformational benefits to organizations.

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Drivers

Demand-side drivers include:

- Increased complexity of software systems to be engineered
- Increased demand for developers to deliver high-quality code faster
- Increased number of quality-related risks associated with software development
- The need to protect against increasingly sophisticated security threats
- The need to optimize operational costs

Supply-side drivers:

- Applying AI models to improve software quality by detecting defects, and fixing them
- Increasing impact of software development on business
- Applying foundation models such as large language models (LLMs) to software code generation and optimization
- Applying deep learning models to software operations

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Obstacles

- The 2024 Gartner Technical Architect Survey highlights a growing skills gap in AI/ML, with 31% reporting high demand but persistent shortages across organizations. Top-performing firms increasingly seek software developers with data science skills. And the 2025 Gartner Software Engineering Survey identifies prompt engineering as an emerging skill, underscoring the rising importance and challenge of addressing these skill shortages.
- Employees fear that job automation by AI will lead to job elimination.
- The hype surrounding this innovation has led to misconceptions and unrealistic expectations regarding AIASE's advantages.
- Myopic focus on code generation has reduced opportunities for AI tool application throughout the SDLC.
- Solutions are uneven and fragmented, automating only certain tasks within the SDLC.
- Intellectual property risks and privacy issues arise from AI models trained on restrictive licensed code and proprietary or leaked data.

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User Recommendations

- Pilot tools to assess potential gains, measure their effectiveness, and then roll out or scale them broadly if the results are positive.
- Innersource best practices, featuring automated prompt generation through saved examples, to effectively use AIASE technology.
- Validate the accuracy and verify the maintainability of AI-generated artifacts via human or automated reviews.
- Enter only short-term contracts with vendors to ensure the ability to migrate to more effective tools as the market matures.
- Ensure psychological safety for software engineers by emphasizing that AIASE serves as a learning partner and augmentation toolset, supporting rather than replacing human expertise, and offering continued career development opportunities.
- Establish guidelines to choose providers that offer transparency in training data and model processes. Prioritize indemnified commercial models for critical applications, and use tools like ScanOSS for licensing compliance.
- Establish the correct set of metrics, such as new release frequency and quality artifacts, to measure AIASE's success.

Sample Vendors

Amazon Web Services; Anima; Atlassian; CAST; Codeium; Dynatrace; GitHub; GitLab; Sedai; Veracode

Gartner Recommended Reading

How Platform Engineering Teams Can Augment DevOps With AI

Market Guide for AI-Augmented Software-Testing Tools

Magic Quadrant for AI Code Assistants

Innovation Insight for AI-Native Software Engineering

Critical Capabilities for AI Code Assistants

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Reinforcement Learning From Human Feedback

Analysis By: Jasleen Kaur Sindhu

Benefit Rating: High

Market Penetration: 5% to 20% of target audience

Maturity: Adolescent

Definition:

Reinforcement learning from human feedback (RLHF) is a machine learning technique that combines automated and human guidance to train an AI agent or a model. It includes training a reward model, based on positive and negative human feedback, then using this model to optimize the AI agent to better understand natural language, learn from human experience and better align with human preference.

Why This Is Important

RLHF is an important technique to steer a self-supervised learning model through a human-based reward mechanism rather than an engineered function. With RLHF, the model does not receive explicit labels or annotations from humans, but instead is externally supervised through signals from interactions with the environment and additional feedback from humans. This improves model performance and accuracy, and helps to align with human preferences, which enables the creation of conversational AI assistants.

Business Impact

RLHF is recognized as a critical technique to ensure that model outputs are truthful and helpful. It has a wide application range from the domain of large language models (LLMs), image generation, gaming and robotics. RLHF-trained LLMs are finding widespread adoption in industries such as education, healthcare, gaming, entertainment and financial services, and in generative AI (GenAI) use cases that allow nontechnical employees and consumers to directly interact with AI.

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Drivers

- New foundation models with advanced capabilities, including multimodal models, are rapidly emerging, making it possible for users to apply LLMs across a range of use cases. Additionally, advancements in agentic AI that can act on behalf of humans is necessitating the need for RLHF.
- RLHF enhances the naturalness and intuitiveness of human-AI interactions by being able to optimize model outputs for desirable user reactions (like satisfaction) and avoiding undesirable ones (like offence).
- Growing oversight and scrutiny of AI implementations, and its impact on customers, employees and society is necessitating the need to improve content appropriateness, and reduce discriminatory or offensive content to support widespread adoption of GenAI-based tools.
- Increasing awareness and concerns about ethical AI development is driving the need for RLHF, which helps align AI operations with human ethical standards and societal values.
- RLHF is beneficial when defining a straightforward, explicit reward function is challenging, as it uses human judgment to guide the model in understanding what constitutes a better response. This approach simplifies the training process by utilizing human feedback, which is more readily available and easier to provide than crafting detailed examples for model training.
- Greater appetite to democratize access to AI-embedded tools and solutions enable nontechnical employees and consumers to directly interact with these interfaces with ease.
- The growing availability of RLHF services from commercial vendors allows firms to adopt and utilize RLHF fine-tune their foundation models.

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Obstacles

- Gathering and processing high-quality human feedback is often time-consuming and costly, limiting the scalability of RLHF.
- Human feedback enhances LLM models' understanding of natural language and aids performance in environments where specifying rewards is challenging. Special attention must be given as RLHF is powerful and can introduce various human biases and errors.
- Accurately capturing complex, fluid and context-dependent human preferences can be difficult, often leading to a misspecified model due to diverse preferences and errors in approximation, estimation and optimization.
- Techniques, such as direct preference optimization, that directly optimize the language model, based on human preference data, could be more cost-effective. Techniques that remove the need for a separate reward model, or reinforcement learning from AI feedback, instead of human feedback, could be used as alternatives.
- LLM models may continue to demonstrate undesirable behaviors that might not be captured by human feedback, or may exploit loopholes in the feedback process to achieve higher rewards.

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User Recommendations

- Avoid using non-RLHF models when the AI must operate in environments where decision making is complex and outcomes are highly variable or unpredictable (e.g., personal assistants in healthcare).
- Apply RLHF-trained models where there is a greater risk of incorrect or inappropriate responses that may negatively impact your business (e.g., a customer-facing application), or use cases with a high risk of biases toward specific customer demographics.
- Consider RLHF for verification and accuracy of information in use cases that require understanding and generation of natural language, such as conversational agents and text summarization.
- Leverage off-the-shelf RLHF models from vendors in the market, but assess the suitability and applicability for industry-specific use cases.
- Conduct regular model assessments and feedback from diverse users to avoid biases. Ensure that the model is effective across various demographics and preferences.

Sample Vendors

Appen; Argilla.io; clickworker; Invisible AI; OpenAI; Prolific; Scale AI; Surge AI; Toloka AI

Gartner Recommended Reading

Tool: Roadmap for Deploying and Managing Generative AI

How to Calculate Business Value and Cost for Generative AI Use Cases

9 Regulatory and Legal Implications of Generative AI

Generative AI Workload Accelerators

Analysis By: Chirag Dekate, Arun Chandrasekaran

Benefit Rating: High

Market Penetration: 20% to 50% of target audience

Maturity: Adolescent

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Definition:

Generative AI (GenAI) workload accelerators — such as graphics processing units (GPUs) or custom application-specific integrated circuits (ASICs) — are chips that operate alongside a CPU and support highly parallel processing operations to develop (train) and run (inference) applications based on large GenAI models.

Why This Is Important

The rapid growth in development of large-scale GenAI models demands significantly more processing performance than can be delivered by standard CPUs. Leveraging workload accelerator chips optimized for these tasks can significantly reduce training times and enable cost-effective deployment of models to support high volumes of user transactions.

Business Impact

Leveraging GenAI workload accelerators enables:

- Reduced time to train or fine-tune large language models (LLMs) used for GenAI for a specific use case or dataset.
- Cost-effective training and deployment of GenAI-based applications for use in a wide spectrum of business use cases.
- Improved total cost of ownership (TCO) and reduced transaction costs when running a wide range of GenAI workloads.

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Drivers

- Training GenAI models requires processing techniques that leverage high-performance systems with workload accelerators.
- Executing GenAI-based applications typically requires computer systems that can execute high volumes of highly parallel math operations, which requires GenAI workload accelerators.
- GenAI models require training with large datasets. Workload accelerators such as data center GPUs can be used for this task, but high-performance custom ASICs are being developed that can deliver a more cost-effective solution to this problem.
- A number of startups that have developed workload accelerator chips will soon bring their products to market, offering broader choices to IT organizations that want to deploy infrastructure to run GenAI-based applications.
- Many GenAI models are being trained by using large cloud-based server clusters that integrate workload accelerators, but some enterprise organizations may need to use workload-accelerator-based infrastructure within their on-premises data centers to deploy these models for use.

Obstacles

- The most commonly used workload accelerator is a high-performance data center GPU. These are expensive and have very high power demands, impacting both server design and the data center infrastructure required to support them.
- Deployments may be limited where a data center has insufficient power per rack or inadequate cooling solutions to support high-power workload accelerators.
- New servers with higher specification CPUs, more memory and storage, along with high-bandwidth networking equipment may be required to deploy GenAI applications in on-premises data centers.
- A limited number of vendors offer ASIC-based workload accelerators that can be used by enterprises as an alternative to GPUs. Limited sourcing creates support, management and supply challenges, further inhibiting adoption of ASICs.
- GPU-based systems are more readily available from a wider range of vendors, and offer support for the broadest set of models. This may limit growth in use of ASIC-based workload accelerators despite their promise of better efficiency and lower power consumption compared to high-performance GPU-based solutions.

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User Recommendations

- Use latest-generation CPUs that integrate accelerators (on-premises or cloud infrastructure) when GenAI workload demand is light enough to fit in conventional CPU-based infrastructure.
- Use GPUs or dedicated workload accelerators when GenAI workloads would otherwise consume excessive server resources or when CPUs can't meet performance and concurrency needs.
- Select workload accelerators and vendors that offer or support the broadest set of GenAI models and toolsets.
- Use compute optimization tools to boost the efficiency of your compute accelerators for better ROI.

Sample Vendors

Amazon Web Services (AWS); AMD; Cerebras; Google; Groq; Intel; Microsoft; NVIDIA; SambaNova Systems; Tenstorrent

Gartner Recommended Reading

Frontier AI Models Like DeepSeek Will Transform I&O Strategies

Forecast Analysis: AI Software Market by Vertical Industry, 2023-2027

Forecast Analysis: AI-Optimized Servers, Worldwide

ModelOps

Analysis By: Cuneyd Kaya, Soyeb Barot

Benefit Rating: Transformational

Market Penetration: 5% to 20% of target audience

Maturity: Adolescent

Definition:

Model operationalization (ModelOps) is primarily focused on the end-to-end governance and life cycle management of advanced analytics, AI and decision models, such as models based on machine learning (ML), generative AI (GenAI) and knowledge graphs.

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Why This Is Important

ModelOps helps companies in standardizing, scaling and augmenting their analytics, AI and GenAI initiatives. It helps organizations to move their models from the lab environments into production. MLOps primarily focuses on monitoring, operationalization and governance of ML models, while ModelOps assists with the same for all advanced analytics, decision and AI models, including GenAI and retrieval-augmented generation (RAG) systems.

Business Impact

ModelOps as a practice:

- Provides the capability for the management and operationalization of diverse AI (including generative AI), analytics and decision intelligence systems.
- Enables the complex subsystems required for such systems with observability, versioning, monitoring, data and model orchestration, guardrails, experimentation and explainability.
- Ensures collaboration among a wider business, development and deployment community and the ability to associate model outcomes with business KPIs.
- Is an important tech capability part of AI engineering; ModelOps' best practices make it possible to deploy models into production in a structured, repeatable factory model.

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Drivers

- Modern AI systems are being built with a symbiotic combination of generative and classic AI models, agents and intelligent software capabilities. As the number of advanced analytics, AI and decision models at organizations increases, organizations will have to manage different types of prepackaged or custom-made models in production. In doing so, it becomes crucial for organizations to ensure these models are working as intended, maintaining accuracy and reliability across various applications.
- Organizations want to be more agile and responsive to changes within their advanced analytics and AI pipelines not only with models, but also with data, application and infrastructure.
- ModelOps provides a framework to separate responsibilities across various teams for how models (including GenAI, foundational models, analytics, ML, physical, simulation and symbolic) are built, tested, deployed and monitored across different environments (for example, development, test and production). This enables better productivity and collaboration and it lowers failure rates.
- ModelOps provides tools to address model degradation via drift and bias. In other scenarios, enabling model runtime enforcements, explainability and integrity is paramount.
- Organizations don't want to deploy an unlimited number of open-source offerings to manage ModelOps, but there are few comprehensive solutions that provide end-to-end capabilities in every domain of model operationalization. Moreover, not every capability is required immediately. Often, versioning, monitoring and model orchestration precede the full implementation of feature stores, pipelines and observability.
- GenAI will require an increased focus on testing and the introduction of capabilities to version, manage and automate prompts, routers and retrieval-augmented generation systems. Fine-tuning will also require enhanced ModelOps capabilities to manage transformations for complex domain and function training datasets.

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Obstacles

- Organizations using different types of models often don't build the right Ops, model usage policy and management capabilities until they already have a chaotic landscape of unmanaged advanced analytic and AI systems.
- Not all analytical techniques currently benefit from mature operationalization methods. Because the spotlight has been on ML techniques, MLOps benefits from a more evolved AI practice, but some models, like agentic modeling and optimization techniques, require more attention in ModelOps practices and platforms.
- Organizations are struggling to get GenAI into production, due to their data not being consumable by AI models, along with data privacy, security and regulatory concerns.
- Organizations may adopt ModelOps platform capabilities that they don't immediately need. At the same time, organizations that are siloed and fail to adopt a comprehensive ModelOps strategy create redundancy in effort with respect to operationalization.

User Recommendations

- Buy ModelOps capabilities integrated into your primary AI platforms. Enrich these capabilities with best-of-breed open-source or proprietary ModelOps offerings where unique problems, like feature stores or observability, require enhanced solutions.
- Utilize ModelOps best practices across composite AI, data, models and applications to ensure transition, reduce friction and increase value generation.
- Recruit/upskill additional engineers who can master ModelOps on AI systems that utilize unstructured data, search and retrieval, graph and optimization.
- Encourage collaboration between data science and development and deployment teams; empower teams to make decisions to automate, scale and bring stability to the analytics and AI pipeline.
- Collaborate with software engineering teams to scale ModelOps. Offloading operationalization responsibilities to production support teams enables increased ModelOps specialization and sophistication across the ecosystem of complex AI-enabled applications.

Sample Vendors

Amazon Web Services; CoreWeave (Weights & Biases); Databricks; Dataiku; DataRobot; Domino Data Labs; Google Cloud; IBM; Microsoft Azure; ModelOp

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Gartner Recommended Reading

Demystify the Ops Landscape to Scale AI Initiatives: A Gartner Trend Insight Report

Launch an Effective Machine Learning Monitoring System

Toolkit: Delivery Metrics for DataOps, Self-Service Analytics, ModelOps and MLOps

Reference Architecture Brief: MLOps Architecture

Critical Capabilities for Data Science and Machine Learning Platforms, AI Engineering

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Sliding into the Trough

Generative AI-Enabled Applications

Analysis By: Radu Miclaus, Arun Chandrasekaran

Benefit Rating: High

Market Penetration: More than 50% of target audience

Maturity: Mature mainstream

Definition:

Generative AI-enabled applications use generative AI models and systems, including virtual assistants and AI agents, for UX and task automation to accelerate and assist the completion of a user's desired outcomes. When embedded in the experience, GenAI offers richer contextualization for tasks like generating and editing text, code, images and other multimodal output. Process-aware GenAI agents can be designed to accelerate workflows that tie multiple tasks together to achieve a desired goal.

Why This Is Important

Fast-moving advances in foundation/specialized models and computational processing enable GenAI-enabled applications that are increasingly feasible and efficient for enterprise use cases. The use cases permeate a wide spectrum of domains, industries and skill sets within the knowledge workforce, reimagining how enterprises think of scale, productivity and assets.

Business Impact

GenAI models, chatbots, agents or AI assistants within applications address time-consuming, manual and repetitive tasks, such as:

- Knowledge discovery
- Summarization and contextualization
- Software engineering and coding
- Graphic and video design
- Business workflow design and execution

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With these tools at their disposal, knowledge workers, creatives and technologists are finding innovative ways to improve employee productivity. Businesses not making use of these tools will struggle to compete.

Drivers

- **Rapid advancement of GenAI models:** Advancements have been fast in foundation (commercial and open source) and specialized (small language and domain-specific) models across multiple dimensions. Models are more approachable for training and refinement and much more efficient in cost and speed for inference, adding to the optionality developers need for building GenAI applications.
- **Wider range of applications:** The most common I/O pattern for GenAI-embedded capabilities today is text-to-X ("X" stands for several modalities). This democratizes access for knowledge workers (via prompt engineering using natural language) to what were once specialized tasks. For example:
  - **Text-to-text** supports knowledge discovery, summarization and contextualization in communication applications across the enterprise.
  - **Text-to-code** is emerging as developer processes get augmented through "pair programming" with AI assistants directly into the coding experience. Use cases range across the SDLC.
  - **Text-to-image/video** (or image-to-image) is the use of GenAI to create images or videos based on text or other images, a capability being integrated into graphic design and video editing applications by both established tech companies and emerging startups.
  - **Text-to-process/workflow** in the form of GenAI agents enables users to use text and voice to generate workflows and generative tasks together in cohesive domain-specific applications.
  - **Text-to-multimodal** supports the building of high-fidelity avatars or digital objects that have image, sound and narrative/text modalities (e.g., multimodal applications in metaverse and gaming). In multimodal work, X-to-text (multiple modality inputs requiring text description and summarization as output) will support ever-growing use cases.
- **Vendor adoption:** Vendors are pushing the pace of adoption of GenAI technologies, offering options for enterprises to build or buy these capabilities.

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- **Computation cost optimization:** The computational innovations for training and inference focus on optimizing and refining the cost structures across the entire software stack (infrastructure, methodologies and integrations), which is a boon for AI agent development.

Obstacles

- **Security, consumer privacy and enterprise IP protection concerns:** Enterprises are concerned with the wide umbrella of trust and security. While large hyperscale vendors and startups are racing to make GenAI services enterprise-ready, in the short to midterm, regulation and appropriate adaptable oversight will lag.
- **Accuracy and veracity of outputs:** Hallucinations and inaccuracy continue to be a concern for GenAI. AI agents require human supervision to ensure they take the proper actions.
- **Cost of training and maintaining generative models:** For direct data and infrastructure cost and indirect costs like sustainability, GenAI model development and refinement continue to be resource-intensive.
- **Regulation:** Globally, regulations are emerging constantly with definitions of prohibited use cases, requirements for disclosure, IP protection, and definitions of risk and mitigation. The divergence in regulatory focus between countries and regions adds an additional burden to developing GenAI applications to be used across jurisdictions.

User Recommendations

- **Seek technology vendors with out-of-the-box GenAI and AI agent applications.** Before building in-house, prioritize offerings available off-the shelf from the growing number of technology vendors with enterprise-grade capabilities.
- **Seek technology providers that offer vertical specialization when building in-house.** Prioritize vendors that accelerate the refinement and adoption of GenAI capabilities and AI agents in vertical and business processes of the enterprise to address existing and future needs.
- **Invest in internal data readiness** to ensure quick integration into GenAI and AI agent grounding processes to produce high-quality, transparent and trustworthy outputs with minimal hallucinations.

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- **Use enterprise-ready** GenAI services for hybrid build-and-buy approaches, with a focus on addressing the privacy, security and IP protection needs of the enterprise.
- **Encourage steady growth** by challenging knowledge workers and technologists to engage in new learning curves, and improve or redesign business and technology processes to respond to this disruption.

Sample Vendors

Adobe; Alibaba; Amazon; Anthropic; Google; Microsoft; OpenAI; Salesforce; ServiceNow; UiPath

Gartner Recommended Reading

A CTO's Guide to the Generative AI Technology Landscape

How to Use Generative AI to Boost Developer Productivity

Quick Answer: Where Can GenAI Impact Information Worker Productivity?

Innovation Insight: Generative AI Chatbot to Improve CX and Agent Productivity

Retrieval-Augmented Generation

Analysis By: Radu Miclaus

Benefit Rating: High

Market Penetration: More than 50% of target audience

Maturity: Early mainstream

Definition:

Retrieval-augmented generation (RAG) is an architecture design pattern that uses search functionality to retrieve relevant data and add it to the prompt of a GenAI model in order to ground the generative output with information. RAG can be used for retrieving both public internet data and data from private organizational knowledge bases.

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Why This Is Important

The majority of available foundation LLMs are trained on general public information. For LLM completions to be contextual and trustworthy for enterprise use cases, they need to be informed by retrieved information from the organizational knowledge bases. The correct RAG implementation reduces hallucinations.

Business Impact

Traditionally, workers spend 20% to 30% of their time looking for information needed to complete business tasks (based on widely reported surveys). RAG can enhance enterprise search and how information is summarized and created, and reduce hallucinations and accelerate productivity. The applications for self-service content consumption for employees and for customer-facing applications have a significant impact on employee and customer satisfaction, brand equity, and workforce productivity.

Drivers

- Searching and finding current information and knowledge documented in enterprises can be frustrating for workers. With RAG, they benefit from a better information retrieval and synthesis approach.
- The advances of GenAI applied toward content summarization and contextualization have enabled search applications to achieve a more robust and focused content consumption experience.
- GenAI service vendors, either via a hyperscale marketplace or via specialty model APIs, are making it easier for organizations to configure the RAG pattern on their knowledge bases. Many more pilot projects are moving to production.
- Competitive pressure to adopt the latest innovations in order to increase productivity and maintain competitive edge drives organizations to use their knowledge bases to support the RAG pattern for both internal- and external-facing applications.
- Availability of technologies for building RAG, as well as RAG being baked in the GenAI assistant experiences, is penetrating enterprises.
- Multiple vendors are offering RAG integrated into their search and research tools; examples include Deep Research from tools like OpenAI ChatGPT, Anthropic Claude, Perplexity and Google NotebookLM. These are driving the adoption of RAG by end users.
- AI agents are increasingly using RAG to get grounded in organizational knowledge to take the correct actions to achieve their intended goals.

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Obstacles

- Underinvestment in enterprise search has been the norm for years. Companies that are still building the discipline and skill sets for robust retrieval and search to support the RAG pattern will experience friction during adoption.
- Configuration of knowledge base access controls for the RAG pattern is not a trivial task and can prevent widespread adoption across all organizational knowledge.
- More complex RAG architectures, like agentic retrieval and knowledge-graph-based retrieval (GraphRAG), are increasingly needed to complement hybrid retrieval (keyword and vector search). This challenge is compounded as enterprises deal with multimodal documents, making implementation more difficult.
- Concerns about IP protection and responsible AI in the use of LLMs are still an obstacle.
- Vendor fragmentation for building complex RAG implementations, as well as maturing enterprise use cases embedded in business workflows, puts a lot of responsibility for engineering and architecture on enterprise technical teams.

User Recommendations

IT and data and analytics leaders looking at adopting GenAI capabilities on top of private and public corporate data should:

- Prioritize RAG investments by assessing the maturity and readiness of the enterprise in relation to knowledge management and quality and information retrieval for employees and customers.
- Pilot applications using the RAG pattern on a well-known knowledge base to assess the lift in the content consumption experience and to gain buy-in for further investment.
- Plan for investment in filling any skills gaps that exist in internal knowledge engineering capabilities, either through upskilling or external hiring.
- Engage with technology vendors or service providers that combine both technology and services to accelerate adoption of vertical or domain-specific RAG architecture, including domain data models, vector database, graph database, LLM vendors and agentic platform vendors.

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Sample Vendors

Amazon Web Services; Coveo; Elastic; Glean; Google; Graphwise; Microsoft; Squirro; Writer

Gartner Recommended Reading

Getting Started With Retrieval-Augmented Generation

Quick Answer: How to Supplement Large Language Models With Internal Data

Prompt Engineering With Enterprise Information for LLMs and GenAI

How to Pilot Generative AI

Vector Databases

Analysis By: Arun Chandrasekaran, Radu Miclaus

Benefit Rating: High

Market Penetration: 5% to 20% of target audience

Maturity: Adolescent

Definition:

Vector databases store numerical representations of data. In such databases, each point is represented by a vector with a fixed number of dimensions, which can be compared via mathematical operations, such as distance measures. Vector databases are commonly used in machine learning (ML) solutions, where vectors represent data features/attributes, such as text embeddings. Storing these vectors in a database enables users to search for similar data points at scale with low latency.

Why This Is Important

Vector databases serve use cases such as similarity search and product recommendation, enabling real-time information retrieval. Emerging approaches in generative AI (GenAI) have made vector databases often part of retrieval-augmented generation (RAG), a popular way of grounding AI with enterprise data. When customers adopt GenAI models, vector databases store the embeddings for fast retrieval.

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Business Impact

Vector databases enable digital business by enhancing semantic search, personalizing recommendations and supporting AI applications with efficient management of unstructured data. They enable deeper insights, scalable data handling and better accuracy of AI, crucial for data-intensive operations. They can be valuable in leveraging data for competitive advantage and in reducing hallucinations, particularly in industries where data volume, complexity and the need for precision in search and analysis are growing exponentially.

Drivers

- **Popularity of vector embeddings in GenAI for grounding large language models (LLMs):** The surge in AI and ML applications across industries necessitates databases that can efficiently handle complex, high-dimensional data. With the rise of AI foundational models, embeddings have become the cornerstone for semantic search and RAG architectures. Furthermore, grounded RAG architectures are part of the emerging AI agent architectures requiring multiple loops through LLMs to gather information, context and complete tasks and workflows. Hence, vector databases are a key for grounding large foundation models and AI agents with enterprise data.
- **Advanced search and personalization needs:** As users demand more personalized experiences and relevant search results, enterprises need databases capable of semantic search and personalization at scale. Vector databases enable searching based on the meaning behind data, rather than just matching keywords, allowing businesses to deliver personalized content and product recommendations, thus enhancing user engagement and satisfaction.
- **Performance and scalability needs:** The applications looking to embed generative methods that use embeddings-based models need back-end services that can respond with low latency to high-concurrency requests (prompts) and responses (completions) for GenAI use cases.
- **Multimodal data handling:** Advanced vector databases now support various data types, including text, images and audio, enabling seamless integration and retrieval across different data modalities.
- **Developer focus:** Developers of new applications are driving the demand for vector databases by presenting use cases that cannot scale without the ability for embeddings to be stored in an optimized structure for high-throughput production applications.

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Obstacles

- Vector databases are superspecialized databases that may cause challenges around data migration, integration and limited extensibility across use cases, although many large database vendors are now starting to offer vector support in existing databases.
- Most vector databases are delivered as cloud-managed services. The complexity of deploying, configuring and operating them outside cloud environments requires deep technical skills and know-how.
- Vector databases can be expensive to implement, given the newness of the technology. Moreover, unlike SQL databases, vector databases lack a universal query language, with different vendors offering varying query formats and optimization techniques.
- For certain use cases dealing with complex information retrieval of various forms of text data or documents that have mixed modalities in the body (text, sections, numbers, tabular data, graphs. etc.), vectorization may not be enough and new approaches like GraphRAG (using graph representations of the entities and relationships in the documents) need to be employed.

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User Recommendations

- Determine whether your functional requirements can be satisfied by incumbent database vendors that are becoming multimodal and can support the storage and retrieval of vector embeddings — you may not always need a purpose-built vector database.
- Prioritize developer experience, ecosystem integration, use-case fit, reliability and performance as important selection criteria, and validate them thoroughly via a proof-of-concept process.
- Select managed, cloud-based vector databases as deployment modes, unless you have stringent requirements and deep technical skills for an on-premises, self-managed deployment mode.
- Adopt testing practices for RAG architectures that allow you to test multiple vector database options when benchmarking performance for specific applications, as vendor-provided benchmarks may not be appropriate for your use cases.
- Conduct internal training and education on the appropriate use cases for vector databases, how to leverage their true potential and effective ways to optimize their deployment and maximize their value.

Sample Vendors

Amazon Web Services (AWS); Elastic; Google; IBM (DataStax); KX; Microsoft; MongoDB; Pinecone; Redis; Weaviate

Gartner Recommended Reading

Innovation Insight: Vector Databases

A CTO's Guide to the Generative AI Technology Landscape

How to Choose an Approach for Deploying Generative AI

Large Language Models

Analysis By: Leinar Ramos, Bern Elliot

Benefit Rating: Transformational

Market Penetration: More than 50% of target audience

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Maturity: Mature mainstream

Definition:

Large language models are AI foundation models that have been trained on vast amounts of unlabeled textual data. Applications can use LLMs to accomplish a wide range of tasks, including question answering, content generation, content summarization, code generation, language translation and conversational chat.

Why This Is Important

LLMs are one of the core technologies powering the GenAI revolution. During the past few years, LLMs have demonstrated surprising and significant capabilities across a wide range of domains and industries. They have served as general-purpose foundation models that can be customized to tackle many different use cases, and they bring state-of-the-art AI capabilities to organizations. They are also increasingly augmented with tools to build broader systems, such as AI assistants and AI agents.

Business Impact

As the most prominent type of foundation model, LLMs are being adopted across all industries and business functions. Organizations are using them to build question-answering systems, summarize and classify documents, translate and edit text, generate and explain programming code, and address many other use cases. LLMs have been rapidly incorporated into many enterprise applications, increasing their reach and impact.

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Drivers

- **Increased compute power:** The exponential increase in computing power and performance over the past decade has enabled the training and use of ever-larger LLMs, improving their performance as they scale.
- **Access to large volumes of unlabeled text:** The public availability of large volumes of textual data, particularly on the web, makes the creation of high-performing LLMs possible.
- **Vendor adoption:** Large technology companies have invested heavily in the development of new LLMs and have applied LLMs to new use cases. Similarly, LLMs are being incorporated into many enterprise applications, expanding adoption of LLMs.
- **Off-the-shelf applications:** LLMs are available through off-the-shelf chatbots, such as ChatGPT, making it easy for business users to get started.
- **Open-weight LLMs:** The improved performance and increased availability of commercially licensed open-weight LLMs are expanding the potential adoption of LLMs and lowering the cost of using them.
- **Growing ecosystem around LLMs:** An expanding ecosystem of tools, plug-ins, extensions, orchestration capabilities and data retrieval layers augments LLMs and facilitates their use, enabling creation of more-complex LLM-based applications.
- **AI agents:** LLMs increasingly serve as the engines of AI agents that can take action more autonomously, expanding the usefulness of LLM-based systems.
- **Ability to customize LLMs:** LLMs can be customized with domain-specific data for specific industries, functions and use cases. This flexibility drives broader application of LLMs in the enterprise.
- **LLM cost reduction:** The training and inference cost of LLMs has been reduced via advanced model architectures, making LLMs more affordable for enterprises.

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Obstacles

- **Unreliability:** LLMs produce inaccuracies and hallucinations, making them less suitable for use cases requiring high accuracy.
- **Lack of explainability:** LLMs are not currently explainable. The need for decision-making transparency will constrain LLM use-case adoption.
- **Multimodality transition:** State-of-the-art foundation models are increasingly multimodal. Thus, LLMs, which are trained only on text, may become less predominant in the future.
- **LLM market complexity:** Organizations struggle to navigate the wide range of LLMs, validate vendor claims and select the right models for their enterprise use cases.
- **Regulatory environment uncertainty:** Intellectual property in LLM training datasets, privacy and confidentiality of enterprise data, and legal liability are some of the issues that will limit adoption.
- **Cybersecurity:** LLMs are susceptible to adversarial attacks and prompt injections that could result in data leakage and malicious LLM use.
- **Energy consumption:** LLM training and use are very energy-intensive. As LLMs become larger and more widely used, their environmental impact is increasing.

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User Recommendations

- Identify AI use cases, such as content generation, knowledge discovery and conversational user interfaces, where LLMs can deliver a more immediate business impact.
- Avoid training LLMs from scratch. Prioritize off-the-shelf applications, or augment LLMs with RAG or fine-tuning instead.
- Follow prompt engineering best practices to optimize LLM performance. Combine LLMs with other applications and tools for more complex design patterns.
- Validate vendor LLM claims. Consider a wide range of criteria when comparing LLMs, including the type of model required, the model's benchmark performance, the ability to fine-tune or customize the model, and the broader ecosystem of tools that can support the model.
- Define a policy for the acceptable use of LLMs and LLM-based applications that clearly identifies the risks of using these tools, as well as the expected user actions to mitigate them.

Sample Vendors

Alibaba Cloud; Anthropic; Cohere; DeepSeek, Google; IBM; Meta; Microsoft; Mistral AI; OpenAI

Gartner Recommended Reading

When Not to Use Generative AI

Innovation Insight: Multimodal AI Explained

Innovation Insight: AI Agents

Synthetic Data

Analysis By: Arun Chandrasekaran, Alys Woodward, Anthony Mullen

Benefit Rating: High

Market Penetration: 5% to 20% of target audience

Maturity: Early mainstream

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Definition:

Synthetic data is a class of data that is artificially generated rather than obtained from direct observations of the real world. Synthetic data is used as a proxy for real data in a wide variety of use cases, including data anonymization, AI and machine learning (ML) development, data sharing, and data monetization.

Why This Is Important

Obtaining and labeling real-world data for AI development is a time-consuming and expensive task. Specifically for use cases such as training models for autonomous vehicles, collecting real data for 100% coverage of edge cases is either practically impossible or incredibly hard. These challenges can be remedied with synthetic data. It can be generated quickly, cost-effectively, and without personally identifiable information (PII) or protected health information (PHI), making it a valuable technology for privacy preservation. The rise of frontier AI models has highlighted synthetic data as a cost-effective means to build scalable models.

Business Impact

Adoption is increasing across various industries, and Gartner predicts a further rise as synthetic data:

- Avoids using PII when training AI models via synthetic variations of original data or synthetic replacement of parts of data.
- Reduces cost and saves time in ML development.
- Improves AI performance as more “fit for purpose” training data leads to better outcomes.
- Enables organizations to pursue new use cases for which minimal real data is available.
- Addresses fairness issues, such as bias and toxicity, more efficiently.
- Enables software testing on realistic yet private data, without legislation risks or in the absence of such data.

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Drivers

- In regulated industries such as healthcare and finance, buyer interest is growing, as synthetic tabular data can be used to preserve privacy in AI training data.
- To meet the increasing demand for synthetic data for natural language automation training, especially for chatbots and speech applications, vendors are bringing new offerings to market, often to train domain generative AI (GenAI) models. This is expanding the vendor landscape and driving synthetic data adoption.
- Synthetic data applications have expanded beyond automotive and computer vision use cases to include data monetization, external analytics support, platform evaluation and the development of test data.
- Transformer and diffusion architectures, the architectural foundations for GenAI, are enabling synthetic data generation at quality and precision levels not seen before. AI simulation techniques are improving synthetic data quality by better recreating real-world representations.
- There is scope for expansion to other data types. While tabular, image, video, text and speech applications are common, R&D labs are expanding the concept of synthetic data to graphs and multimodal AI. Synthetically generated graphs will resemble but not overlap the original. As organizations begin to use graph technology more, we expect this method to mature and drive adoption.
- As data providers for training frontier AI models raise their data access costs, synthetic data is gaining traction as an economic alternative.

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Obstacles

- Synthetic data can have implicit bias problems, miss natural anomalies, be complicated to develop, or not contribute any new information to existing, real-world data.
- While synthetic data reduces privacy risks, some industries (e.g., healthcare, finance) still face regulatory uncertainty about whether it can fully replace or augment real data for compliance purposes.
- Synthetic data generation methodologies lack standardization.
- Validating the accuracy of synthetic data is difficult. It can be challenging to know for sure whether a synthetic dataset accurately captures the underlying real-world environment.
- Buyers are still confused about when and how to use the technology due to the lack of skills.
- Enterprises have legacy data pipelines, and integrating synthetic data into existing data lakes, analytics systems and AI workflows often requires additional effort, tooling and infrastructure.
- There may be a level of user skepticism as data may be perceived to be “inferior” or “fake.”

User Recommendations

- Identify areas in your organization where data is missing, incomplete or expensive to obtain, and is thus, currently blocking AI initiatives.
- Establish clear policies on synthetic data life cycle management, storage and access control.
- Work with legal and compliance teams to ensure synthetic data aligns with industry regulations, such as General Data Protection Regulation, Health Insurance Portability and Accountability Act and the California Consumer Privacy Act.
- Educate internal stakeholders through training programs on the benefits and limitations of synthetic data. Institute guardrails to mitigate challenges such as user skepticism and inadequate data validation.
- Measure and communicate the business value and the success and failure stories of synthetic data initiatives.

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Sample Vendors

Anyverse; Betterdata; Bifrost; Gretel; MOSTLY AI; Parallel Domain; Rendered.ai; SAS; Tonic.ai; YData

Gartner Recommended Reading

Innovation Guide for Generative AI Models

Executive Briefing on Emerging Technology: Synthetic Data

Market Guide for Data Masking and Synthetic Data

GenAI Virtual Assistants

Analysis By: Danielle Casey, Bern Elliot

Benefit Rating: High

Market Penetration: 20% to 50% of target audience

Maturity: Early mainstream

Definition:

GenAI virtual assistants (VAs) represent a new generation of VAs that leverage LLMs to deliver functionality not obtained with traditional conversational AI technology. GenAI has advanced Q&A support, supported generation features and modalities, and improved task automation and value outcomes.

Why This Is Important

LLMs have materially augmented VAs, with virtually all VA providers having already productized GenAI capabilities. LLMs and multimodal models can be embedded into a VA or can be accessed via an API call. There are multiple approaches to using an LLM — deployed out-of-the box, fine-tuned or retrained for customization, chained with additional LLMs across various tasks, or contextualized via prompt engineering. Each option must be evaluated against performance requirements, time and costs.

Business Impact

GenAI-enabled VAs:

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- Operate with improved performance in conversational dialogue, enabling greater contextuality, reducing development effort, improved self-service and reduced operational support costs.
- Extend automation by expanding from Q&A support to content discovery and creation features.
- Improve user interactions and engagement by supporting multimodal interactions (beyond text and voice; for example, via images and videos).

Drivers

- **Demand for GenAI value:** GenAI-enabled VAs are a popular use case, as the use cases and KPIs of VA implementations are well-defined and measurable, and the need to improve customer or employee support may already be recognized. This makes VAs a good option for organizations piloting GenAI.
- **GenAI-enabled use cases:** Organizations investing in GenAI VAs are looking for promised value outcomes, such as productivity improvement and cost reduction. Some of the associated use cases are Q&A support for customers, information discovery and generation features for employees and intelligent agent assist for call center operators. GenAI VAs are also becoming more specialized and offering deeper automation by use case, business role or industry domain.
- **Rapid time to market:** GenAI VA offerings are focused on lowering the barrier to adoption via easy-to-use platforms where users can select their preferred LLMs, prepackaged use case integrations, etc. These platforms often lower the technical skills barrier by supporting the use of natural language prompts to create GenAI VA flows.
- **AI agent hype:** The emergence of AI agents has reengaged interest with GenAI VAs, as many providers adjust their marketing to include “agent” terminology. Moreover, domain specialized models, large action models and multimodal architectures are advancing GenAI VAs to tackle more complex tasks and use cases. Most VA providers are working on advancing their offerings to support agentic automation, though adoption of AI agents is emerging.

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Obstacles

- **Explainability:** Explainable GenAI outputs are key to addressing accuracy and reliability concerns. Providers are working to embed outcome explainability and performance reliability — often via combining GenAI with rule-based tech — but this remains an unsolved problem.
- **Customization requirements:** To unlock value, GenAI VAs must be connected to customer data via prompt engineering, retrieval-augmented generation, model fine-tuning or other techniques, requiring customer education, budget, skills and data readiness.
- **Solution performance:** Many GenAI VAs struggle to deliver strong value outcomes using general purpose LLMs. Addressing this challenge requires significant model or data customization, such as fine-tuning LMs for specific domains or use cases. Domain-specialized models will also help here, but remain scarce.
- **Privacy and security:** Privacy and security concerns may necessitate stricter model hosting and data integration requirements.
- **ROI:** Tokenization and generative inferences are changing VAs' cost structure. Difficulty in proving ROI may slow adoption.

User Recommendations

- Define and prioritize use cases. GenAI VAs offer low-risk, measurable use cases rendering it ideal for piloting and scaling GenAI within the organization.
- Consider solutions that offer hybrid options — both LLM and rule-based flows — for better cost-effectiveness, responsiveness, explainability and robustness.
- Measure implementation success by clearly identifying the business case and supported value outcomes. Regularly assess performance to identify model accuracy issues or business misalignment.
- Improve GenAI VA performance and drive deeper automation by customizing your VAs using model fine-tuning, advanced RAG techniques or process-engineered workflow automation. This will improve VA knowledge support for particular use cases, business roles or vertical industries.
- Prepare for agentic automation. Start with GenAI VAs before progressing to agentic automation and the technology, business and change management challenges this emerging tech introduces.

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Sample Vendors

Aisera; Amelia; Google; IBM; Kore.ai; Leena AI; OneReach.ai; Openstream.ai; OrbyAI; Salesforce

Gartner Recommended Reading

Emerging Tech: Market Risk Projection of Generative AI on Conversational AI

Market Guide for Conversational AI Solutions

Emerging Tech: Conversational AI Differentiation in the Era of Generative AI

Appendixes

See the previous Hype Cycle: Hype Cycle for Generative AI, 2024.

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Hype Cycle Phases, Benefit Ratings and Maturity Levels

Table 2: Hype Cycle Phases

(Enlarged table in Appendix)

Phase	Definition
Innovation Trigger	A breakthrough, public demonstration, product launch or other event generates significant media and industry interest.
Peak of Inflated Expectations	During this phase of overenthusiasm and unrealistic projections, a flurry of well-publicized activity by technology leaders results in some successes, but more failures, as the innovation is pushed to its limits. The only enterprises making money are conference organizers and content publishers.
Trough of Disillusionment	Because the innovation does not live up to its overinflated expectations, it rapidly becomes unfashionable. Media interest wanes, except for a few cautionary tales.
Slope of Enlightenment	Focused experimentation and solid hard work by an increasingly diverse range of organizations lead to a true understanding of the innovation's applicability, risks and benefits. Commercial off-the-shelf methodologies and tools ease the development process.
Plateau of Productivity	The real-world benefits of the innovation are demonstrated and accepted. Tools and methodologies are increasingly stable as they enter their second and third generations. Growing numbers of organizations feel comfortable with the reduced level of risk; the rapid growth phase of adoption begins. Approximately 20% of the technology's target audience has adopted or is adopting the technology as it enters this phase.
Years to Mainstream Adoption	The time required for the innovation to reach the Plateau of Productivity.

Source: Gartner (July 2025)

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Table 3: Benefit Ratings

Benefit Rating	Definition
Transformational	Enables new ways of doing business across industries that will result in major shifts in industry dynamics
High	Enables new ways of performing horizontal or vertical processes that will result in significantly increased revenue or cost savings for an enterprise
Moderate	Provides incremental improvements to established processes that will result in increased revenue or cost savings for an enterprise
Low	Slightly improves processes (for example, improved user experience) that will be difficult to translate into increased revenue or cost savings

Source: Gartner (July 2025)

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Table 4: Maturity Levels

(Enlarged table in Appendix)

Maturity Levels	Status	Products/Vendors
Embryonic	In labs	None
Emerging	Commercialization by vendors Pilots and deployments by industry leaders	First generation High price Much customization
Adolescent	Maturing technology capabilities and process understanding Uptake beyond early adopters	Second generation Less customization
Early mainstream	Proven technology Vendors, technology and adoption rapidly evolving	Third generation More out-of-box methodologies
Mature mainstream	Robust technology Not much evolution in vendors or technology	Several dominant vendors
Legacy	Not appropriate for new developments Cost of migration constrains replacement	Maintenance revenue focus
Obsolete	Rarely used	Used/resale market only

Source: Gartner (July 2025)

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Acronym Key and Glossary Terms

AI	artificial intelligence
API	application programming interface
CTO	chief technology officer
GenAI	generative artificial intelligence
GPT	generative pretrained transformer
IoT	Internet of Things
LLM	large language model
ModelOps	model operationalization
POC	proof of concept
RAG	retrieval-augmented generation
SaaS	software as a service
TRiSM	trust, risk and security management

Document Revision History

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Hype Cycle for Generative AI, 2023 - 11 September 2023

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

Understanding Gartner's Hype Cycles

Tool: Create Your Own Hype Cycle With Gartner's Hype Cycle Builder

Innovation Guide for Generative AI Technologies

Demystifying Generative AI Deployment Approaches: Critical Trade-Offs for CIOs to Navigate

Innovation Insight for the AI Agent Platform Landscape

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Table 1: Priority Matrix for Generative AI, 2025

Benefit	*Years to Mainstream Adoption*			
↓	***Less Than 2 Years*** ↓	***2 to 5 Years*** ↓	***5 to 10 Years*** ↓	***More Than 10 Years*** ↓
Transformational	Agentic AI	AI Agent Development Frameworks AI-Augmented Software Engineering AI Supercomputing Embodied AI Large Language Models Large Reasoning Model ModelOps Multimodal Generative AI Self-Supervised Learning	World Models	Artificial General Intelligence

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Benefit	*Years to Mainstream Adoption*			
↓	***Less Than 2 Years*** ↓	***2 to 5 Years*** ↓	***5 to 10 Years*** ↓	***More Than 10 Years*** ↓
High	GenAI Virtual Assistants Generative AI-Enabled Applications Generative AI Workload Accelerators Open-Source LLMs Retrieval-Augmented Generation	AI TRiSM Domain-Specific GenAI Models Edge GenAI Embedding Models GenAI Application Orchestration Frameworks Prompt Engineering Reinforcement Learning From Human Feedback Synthetic Data Vector Databases	Disinformation Security Model Context Protocol	
Moderate	GraphRAG		AI Gateways	
Low				

Source: Gartner (July 2025)

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Table 2: Hype Cycle Phases

Phase	Definition
Innovation Trigger	A breakthrough, public demonstration, product launch or other event generates significant media and industry interest.
Peak of Inflated Expectations	During this phase of overenthusiasm and unrealistic projections, a flurry of well-publicized activity by technology leaders results in some successes, but more failures, as the innovation is pushed to its limits. The only enterprises making money are conference organizers and content publishers.
Trough of Disillusionment	Because the innovation does not live up to its overinflated expectations, it rapidly becomes unfashionable. Media interest wanes, except for a few cautionary tales.
Slope of Enlightenment	Focused experimentation and solid hard work by an increasingly diverse range of organizations lead to a true understanding of the innovation's applicability, risks and benefits. Commercial off-the-shelf methodologies and tools ease the development process.
Plateau of Productivity	The real-world benefits of the innovation are demonstrated and accepted. Tools and methodologies are increasingly stable as they enter their second and third generations. Growing numbers of organizations feel comfortable with the reduced level of risk; the rapid growth phase of adoption begins. Approximately 20% of the technology's target audience has adopted or is adopting the technology as it enters this phase.
Years to Mainstream Adoption	The time required for the innovation to reach the Plateau of Productivity.

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Source: Gartner (July 2025)

Table 3: Benefit Ratings

Benefit Rating	Definition
Transformational	Enables new ways of doing business across industries that will result in major shifts in industry dynamics
High	Enables new ways of performing horizontal or vertical processes that will result in significantly increased revenue or cost savings for an enterprise
Moderate	Provides incremental improvements to established processes that will result in increased revenue or cost savings for an enterprise
Low	Slightly improves processes (for example, improved user experience) that will be difficult to translate into increased revenue or cost savings

Source: Gartner (July 2025)

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Table 4: Maturity Levels

Maturity Levels	Status	Products/Vendors
Embryonic	In labs	None
Emerging	Commercialization by vendors Pilots and deployments by industry leaders	First generation High price Much customization
Adolescent	Maturing technology capabilities and process understanding Uptake beyond early adopters	Second generation Less customization
Early mainstream	Proven technology Vendors, technology and adoption rapidly evolving	Third generation More out-of-box methodologies
Mature mainstream	Robust technology Not much evolution in vendors or technology	Several dominant vendors
Legacy	Not appropriate for new developments Cost of migration constrains replacement	Maintenance revenue focus
Obsolete	Rarely used	Used/resale market only

Source: Gartner (July 2025)

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